

REASONING & COMPUTER APTITUDE**1. Statements:**

- I. Despite the acknowledgment of the lack of power in schools by the Parliamentary Standing Committee, which operates its own schemes and sponsors various school education programmes, a significant number of government schools in India remain unelectrified.
- II. Data indicates that only 56.45% of government schools have electricity, 56.98% have a playground, and almost 40% lack a boundary wall.

Conclusions:

- I. The slow progress in providing such essential facilities, despite sanctioned projects, highlights the low priority given to school education.
- II. The government-run school sector requires a significant infusion of funds, and prioritizing a public school system with universal access, quality learning, and adequate facilities must be among the nation's highest goals.
- (a) Only conclusion I follows
(b) Only conclusion II follows
(c) Either conclusion I or II follows
(d) Neither conclusion I nor II follows
(e) Both conclusions I and II follow
2. Read the statement below carefully and answer the question that follows:
- The introduction of the goods and Services Tax (GST) in July 2017 and the government's demonetization policy have stressed the SME sector. Additionally, rising interest rates for loans against property (LAP) extended to small and medium-sized enterprises (SMEs) are expected to increase delinquencies. This combination of factors will further challenge SMEs in the muted operating environment of 2018, exacerbating risks for both borrowers and lenders.
- Which of the following can be logically inferred from the statement above?
- (a) The increasing delinquencies over the years will force lenders to adopt stricter underwriting practices. This could limit refinancing options, adversely affecting existing borrowers.
(b) Losses on LAP extended to SMEs will be limited because these loans are secured and have relatively low loan-to-value ratios.
(c) Rising interest rates will limit refinancing options and increase repayment amounts, heightening risks for asset-backed securities (ABS) supported by LAP to SMEs.
(d) The ABS backed by LAP extended to SMEs have features like non-amortizing cash reserves, substantial excess spread, and the ability to extend the loan tenure, reducing risks for lenders.
(e) None of the above can be inferred.

Directions (3-5): Study the following information carefully and answer the below questions.

There are four junctions - E, F, G and H. E is to the north of F and G is to the east of H such that the lines joining EF intersect GH at point K. E%P means Car P starts from E, reaches point K then turns to its left. E@Q means Car Q starts from E, reaches point K then turns to its right.

H#R means Car R starts from H, reaches point K then goes straight.

F*S means Car S starts from F, reaches point K then takes U turn.

H%O; G@T; F@Z; E#Y; G#X; F*M

3. Which of the following cars reaches, Junction E finally?
(a) T (b) O (c) X
(d) Both (a) and (b) (e) Both (b) and (c)
4. Four of the following five are alike in a certain way and hence form a group. Which of the following one does not belong to that group?
(a) ET (b) GZ (c) HX
(d) FM (e) GO
5. What is the direction of Z with respect to Y after the final movement?
(a) North-east (b) North-west
(c) South-east (d) South-west
(e) Can't be determined

Directions (6-10): Study the following information carefully and answer the below questions.

Twelve buildings are arranged from north to south in two rows. Each row has 6 buildings. In row 1 - A, B, C, D, E, and F are facing towards east while in row 2 - M, N, O, K, L and H are facing towards west. The buildings in both rows are exactly facing towards each other. Each building has different number of offices. None of the buildings has more than 25 offices. F is located third to the left of the building which has 9 offices. The number of buildings located to the left of F is the same as the number of buildings located to the right of E. H is located exactly opposite to E. Only two buildings are located between H and the building which has 17 offices. N is located third to the left of M. Both M and H are not located adjacent to each other. C is located exactly opposite to the building which is located to the immediate right of M. K is located third to the left of the building which has 12 offices. N has twice the offices of L. A is located fourth to the left of D. F has one office less than that of D. E has twice the offices of F. The sum of the number of offices of O and L is two more than the sum of the number of offices of E and C. A has three offices more than that of C. M has two offices less than that of O. The sum of the number of offices of B and K is one less than that of N. H has twice the offices of K. B has one office less than that of A.

6. Which of the following building is located second to the right of the building which has 24 offices?
(a) The building which has 12 offices
(b) The building which has 15 offices
(c) K (d) O (e) H
7. Which of the following pair of buildings has an even number of offices?
I. LH II. AF III. BD
(a) Only I and III (b) Only I
(c) Only I and II (d) Only II
(e) Only III
8. What is the average number of offices in buildings L, K, and C?
(a) 12 (b) 11 (c) 10
(d) 9 (e) 14
9. Which of the following statement(s) is/are true with respect to the final arrangement?
(a) M has 15 offices
(b) The sum of the number of offices of H and K is 30
(c) A is located to the immediate left of F
(d) Only two buildings are there between O and H
(e) All the above statements are true

10. How many buildings have more than 15 offices?

- (a) None (b) Three (c) Two
(d) Four (e) One

Directions (11-15): Study the following information carefully and answer the below questions.

Seven persons - A, B, C, D, K, L and N are sitting in a linear row such that all of them are facing towards north, but not necessarily in the same order. They are having different years of experience and each of them has at least one year of experience.

Note: If the person has an even number of experience, then the immediate neighbors of that person must have an odd number of experience and vice versa.

D sits third to the left of the one who has 21 years of experience. N sits third to the left of the one who has 8 years of experience. The number of persons sitting to the left of the one who has 13 years of experience is the same as the number of persons sitting to the right of C. B sits to the immediate right of the one who has 24 years of experience. The persons who sit at the ends of the row are not having more than 20 years of experience. The sum of the experiences of N and B is six years more than the experience of D. L has three years less than that of C. L sits second to the right of K. The sum of the experiences of A and K is one year more than the experience of C.

11. As many persons sitting to the left of N as to the right of _____?

- (a) D
(b) L
(c) The one who has 21 years of experience
(d) The one who has 17 years of experience
(e) A

12. Who among the following person sits between N and K?

- I. D
II. B
III. The one who has 21 years of experience
IV. The one who has 14 years of experience
(a) Only I and II (b) Only III and IV
(c) Only I and III (d) Only II and IV
(e) All I, II, III and IV

13. Who among the following person sits at the right end of the row?

- (a) C
(b) L
(c) B
(d) The one who has 17 years of experience
(e) The one who has 13 years of experience

14. How many people sit between B and the one who has 14 years of experience?

- (a) No one (b) Four (c) Three
(d) One (e) Two

15. What is the position of N with respect to the one who has 17 years of experience?

- (a) Fourth to the right (b) Third to the left
(c) Third to the right (d) Second to the left
(e) Second to the right

Directions (16-20): Study the following information carefully and answer the below questions.

Nine persons - M, N, O, P, Q, R, S, T and U are working in three different departments - Biotech, Electronic, and Mechanical and they like different colors - Silver, Navy, Teal, Golden, Blue, Pink, Green, Maroon, and White. All the information is not necessarily in the same order. Note: At least two persons and not more than four persons are working in the same department. If it is given that "M is working with the one who likes Silver", then it means M does not like Silver but another person working along with M in the same department likes Silver. S works with only the one who likes Green. S does not work in the Electronic Department. Both M and Q work in the same department. N works with the one who likes

Maroon. Both O and R work in the same department along with N. R does not like Maroon. T works with the one who likes Blue. Both M and T work in the same department. Neither M nor Q likes Blue. P neither works in Electronic nor Mechanical department. U does not like Blue. R works with the one who likes White. Both R and T like Golden and Teal colors, respectively. The one who likes pink does not work with Q. Q does not like Silver.

16. Which of the following combination(s) is/are true with respect to the final arrangement?

- I. Biotech-N
II. O-Golden
III. S-Pink
(a) Only I and III (b) Only III
(c) Only I and II (d) Only II
(e) None of the statements is false

17. Which of the following color is liked by the one who works with U?

- (a) Teal (b) White (c) Golden
(d) Pink (e) Maroon

18. Which of the following pair of persons are working in the same department?

- I. U and the one who likes Navy
II. The one who likes White and the one who likes Maroon
III. The one who likes Silver and O
(a) Only II (b) Only I and III
(c) Only III (d) All I, II, and III
(e) None of these

19. Four of the following five are alike in a certain way and hence form a group, who among the following one doesn't belong to the group?

- (a) M -Navy (b) Q -Teal (c) T -Green
(d) N -Golden (e) U -Pink

20. Who among the following person works in the Mechanical department?

- (a) The one who likes Silver
(b) The one who likes Green
(c) The one who likes Maroon
(d) R
(e) P

Directions (21-23): Study the following information carefully and answer the question given below:

There are two rectangular parks such that one park is inscribed into another park. Eight gates i.e. P, Q, R, S, T, U and V are there in both the parks, but not necessarily in the same order. Each park has four gates. The gates of the inner park are open towards the outside whereas the gates of the outer park are open towards the inside. All the gates are in the middle of the sides of the parks. The gates of the inner park are facing exactly the gates of the outer park.

Note: Only the gates of different parks are facing towards each other.

Gate P is on the longer side of the inner park. Only two gates are between gates P and T. Gate V is facing towards gate S. Only one gate is between gates U and S. Gate R is facing the gate which is second to the right of gate W. Gate R is not adjacent to gate S. The number of gates between P and R is the same as the number of gates between Q and W. Gate R is not on the immediate right of gate T.

21. Which among the following statement(s) is/are true with respect to the final arrangement?

- I. Both gates R and P open towards inside.
II. Gate R is facing the gate which is to the immediate right of gate U.
III. Both the gates W and S are in the same park.
IV. The number of gates between W and S is the same as the number of gates between V and T.

- (a) Only I (b) Both II and III
(c) All are true (d) Both III and IV
(e) Only II

22. Four of the following five are related in a certain way and hence form a group. Find the following one which does not belong to that group?

- (a) SU (b) QW (c) PR
(d) VT (e) PW

23. Which of the following gates are facing towards each other?

- (a) QP (b) VU (c) TW
(d) SW (e) QR

Directions (24-28): Study the following information carefully and answer the below questions.

Five couples are living in a five-storey building such that the bottommost floor is numbered as one and the topmost floor is numbered as five.

- I. Each floor has 2 flats viz., flat X and flat Y. Flat X is to the west of Flat Y.
 - II. Flat X of floor 2 is immediately above flat X of floor 1 and immediately below flat X of Floor3 and so on.
 - III. The person whose name starts with a vowel is a female member. The couples are living on the same floor of the building. The number of floors above Arnab is one less than the number of floors below Umesh. Olivia's spouse lives immediately below Umesh in the same flat. Iman's spouse lives to the east of Iman, who lives above Gita. Mita lives to the south of Pinaki's spouse. Bablu lives northeast of Umesh's spouse.
24. Who among the following person lives on the same floor of Iman?
- (a) Pinaki
(b) The one who lives immediately below Chandra
(c) Rahul
(d) The one who lives immediately above Umesh.
(e) Gita
25. If Pinaki is related to Mita and Rahul is related to Bablu in a certain way. Then who among the following is related to Gita?
- (a) Umesh's spouse
(b) The one who lives immediately above Rahul
(c) Arnab
(d) Mita
(e) Bablu
26. What is the direction of Rahul's spouse with respect to Bablu's spouse?
- (a) North-east (b) South-east
(c) South-west (d) North-west
(e) North
27. Who among the following person lives in flat Y?
- (a) Pinaki
(b) Rahul
(c) Iman
(d) The one who lives on the same floor with Gita
(e) The one who is the spouse of Iman
28. On which of the following floor and flat does Gita's spouse live?
- (a) Floor 4, Flat Y (b) Floor 3, Flat Y
(c) Floor 2, Flat X (d) Floor 1, Flat Y
(e) Floor 1, Flat X

Directions (29-30): Each of the questions below consists of a question and three statements numbered I, II, and III are given below it. You have to decide whether the data provided in the statements are sufficient to answer the question:

29. Is Q the grandmother of N?

Statements:

- I. P is the father of O, who is the mother of N. K is the brother-in-law of S.

II. S is the father of R, who is the brother of N. Q is the mother-in-law of O's spouse.

III. K is the Uncle of R. S does not have any siblings.

- (a) Both Statements I and II together is sufficient to answer the question
(b) Both statements II and III together is sufficient to answer the question
(c) Both statements I and III together is sufficient to answer the question
(d) Only statement II alone is sufficient to answer the question
(e) None of the above
30. Seven persons - A, B, C, D, E, F and G are sitting on the circular table such that all of them are facing towards the center, but not necessarily in the same order. Who among the following person sits third to the right of A?

Statements:

- I. D sits second to the right of G, who sits to the immediate left of A. One person sits between A and B, who doesn't sit adjacent to G.
II. F sits second to the left of E. Two persons sit between E and A.
III. C sits to the immediate right of F. Both A and F are not immediate neighbors.
- (a) Both Statements I and II together is sufficient to answer the question
(b) Both statements II and III together is sufficient to answer the question
(c) Both statements I and III together is sufficient to answer the question
(d) Only statement II alone is sufficient to answer the question
(e) None of the above

Directions (31-34): Study the following information carefully and answer the below questions.

A word and number arrangement machine, when given an input line, rearranges them by following a particular rule in each step. The following is an illustration of input and the rearrangement.

Input: Across Day 06 Eager Bear 23 31 However Informal 17 12 14

Step I: Day 07 Across Eager Bear 23 31 However Informal 17 12 14

Step II: Day 07 Bear 13 Across Eager 23 31 However Informal 17 14

Step III: Day 07 Bear 13 Eager 15 Across 23 31 However Informal 17

Step IV: Day 07 Bear 13 Eager 15 Across 16 23 31 However Informal

Step V: Day 07 Bear 13 Eager 15 Across 16 However 22 31 Informal

Step VI: Day 07 Bear 13 Eager 15 Across 16 However 22 Informal 30

Step VI is the last and the final output of the input. By following the same rules, find the steps and the final output of the input given below.

Input: Identify Eat 43 16 Actual 24 li ke 37 20 Gears However 39

31. How many elements are there between "Gears" and "37" in step III of the given input?
- (a) Five (b) Four (c) Three
(d) Six (e) Two
32. What is the sum of the last three numbers from the right end in step VI of the given input?
- (a) 107 (b) 108 (c) 113
(d) 116 (e) 112
33. Which of the following represents step IV of the given input?
- (a) Eat 17 like 21 Identify 41 Actual 26 35 Gears However 39
(b) Eat 17 like 21 Gears 27 Identify 41 Actual 35 However 39
(c) Eat 17 like 21 Gears 25 Actual 36 Identify 43 However 39
(d) Eat 17 like 21 Gears 27 Actual 36 Identify 36 41 However
(e) Eat 17 like 21 Identify 41 26 Actual 36 Gears However 39
34. As many elements are arranged to the left of "Like" in step IV as to the right of _____ in step II of the given input?
- (a) 37 (b) However (c) Gears
(d) 43 (e) Actual

Directions (35 - 37): In the following questions, the symbols @, \$, and & are used with the following meanings as illustrated below. In each of the questions, some statements are followed by some conclusions. You have to take the given statements to be true and disregard the commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follow from the given statements and disregard the commonly known facts.

P@Q → Only a few P is Q

P\$Q → No P is Q

P&Q → Some P is Q.

35. Statements:

G@X&E\$D

Conclusions:

I. Some X are not D.

II. No E is G.

III. Some E can be G.

IV. All X can never be E.

(a) Both II and III (b) Both III and IV

(c) Both I and III (d) Both I and II

(e) All are true

36. Statements:

A@H\$O&M

Conclusions:

I. Some M are A.

II. All A can never be O.

III. No A is O.

IV. Some M can be H.

(a) Both II and III (b) Both II and IV

(c) Both I and III (d) Both I and II

(e) All are true

37. Statements:

I@M&X&B

Conclusions:

I. Some M can be B.

II. Some I are not M.

III. Some X being B is a possibility.

IV. No X is I

(a) Both II and III (b) Both II and IV

(c) Both I and III (d) Both I and II

(e) All are true

Directions (38-42): Study the following information carefully and answer the below questions.

The management arranged the guest lecture in the auditorium. Some students are attending the guest lecture and they are sitting in three different rows such that all of them are facing towards north. Row1 is to the north of Row2 and Row3 is to the south of Row2. Each row has 10 seats. The right-most end seats of three rows, as well as the first seat from the left end of row2 and row3, are restricted. Twelve students - A, B, D, G, J, L, N, O, R, T, V, and Y forming two different groups - Group A and Group B are attending the guest lecture, but not necessarily in the same order.

Group A: The students whose name starts with a letter that comes before M in the English alphabetical series belong to this Group.

Group B: The students whose name starts with a letter that comes after M in the English alphabetical series belong to this Group.

The students in group A are sitting in the alternative seats starting with the second seat from the left end of Row 1 till Row3 in the reverse alphabetical order followed by the group B students in the alphabetical order. All students are sitting in the alternative seats only.

Then, all the students are seated at two different circular tables - Table1 and Table2 such that all of them are facing towards the center. Six students are sitting on each table.

Table 1: J sits second to the left of the one who sits to the immediate right of A in Row-2. The one who sits to the immediate right of J in row1 sits third to the left of the one who sits second to the left of Y in Row-3. The one who sits to the immediate right of N in row-2 sits to the immediate left of G. Both the one who sits to the immediate left of T in row-3 and G are immediate neighbors.

Table 2: Y sits third to the left of the one who sits to the immediate right of G in row-1. A sits second to the right of the one who sits to the immediate right of G in row-1. The one who sits to the immediate left of Y in row-3 sits to the immediate left of the one who sits second to the right of Y. The one who sits to the immediate left of J in row-1 sits second to the right of the one who sits to the immediate left of A in row-2.

38. Who among the following students does not sit on row2?

(a) B

(b) The one who sits to the left of J

(c) A

(d) The one who sits to the immediate left of O

(e) The one who sits to the immediate right of N

39. Who among the following students sits third to the right of T in Table1?

(a) The one who sits to the immediate left of R in table 1

(b) The one who sits second to the right of O in table 1

(c) J

(d) N

(e) None of the above

40. Who among the following students sits in table 2?

(a) G

(b) R

(c) The one who sits to the immediate left of B.

(d) O

(e) N

41. What is the position of D with respect to L in Table 2?

(a) Third to the left

(b) Second to the right

(c) Second to the left

(d) Immediate right

(e) Immediate left

42. Which of the following statement(s) is/are true with respect to the given conditions?

(a) Both T and N are not immediate neighbors in Table 1

(b) V sits second to the left of D in Table 2

(c) Both T and Y sit in the different rows

(d) Both A and D are not sitting at the same table

(e) All the statements are true

Directions (43-44): In these questions, certain symbols have been used to indicate relationships between elements as follows. Find the conclusion which is definitely true.

"A @ B" means "A is greater than B"

"A % B" means "A is not greater than B"

"A # B" means "A is less than B"

"A & B" means "A is not less than B"

"A ^ B" means "A is neither greater than nor less than B"

43. Statements:

A#C%B^O;

Y@S&N^E^O

Conclusions:

(a) C@N (b) A%S

(c) S@O

(d) C#E (e) C%S

44. Statements:

A@V&U&B;

K#C%B@P

Conclusions:

(a) A^C (b) V&C

(c) P@U

(d) K&P (e) A&P

45. A certain number of persons are standing in the queue one after another.

At least 9 persons and not more than 12 persons are standing in the queue. M stands immediately in front of O. Only three persons stand between O and N. P stands immediately after N who does not stand before M. Six persons stand between P and F. A stands five persons before B. Only two persons stand between P and B.

How many persons stand between F and A?

- (a) Eight (b) Nine (c) Three
(d) Six (e) Five

ENGLISH LANGUAGE

Directions (46-50): Read the following passage and answer the questions given below. Some words are highlighted to help you answer some of the questions.

We are now living among history's very first "AI" generation. From the Alexas they converse with, to their robot playmates, to the YouTube wormholes they disappear into, the children and adolescents of today are born into a world increasingly powered by virtual reality and artificial intelligence (AI). AI is not only **(I) revamping** what humans can do, it is shaping our behaviors, our preferences, our perceptions of the world and of ourselves. Older people still remember life before AI and the digital world - our references, anchors and pole stars pre-date the fourth Industrial Revolution. Not so for the millions of children and adolescents who were born into it. One of the most **pressing** concerns is that not everyone can tap into the opportunities offered by this transformation. According to UNICEF and the International Telecommunication Union (ITU), as many as two-thirds of the world's children do not have access to the Internet at home.

In the old-fashioned physical world, we evolved norms and standards to protect children. For instance, there are policies and protocols for a child traveling alone as an unaccompanied minor. Parents are understandably **(II)** to let their children be photographed by the media, and in many countries, news outlets blur children's faces to protect them. While video gaming and chat forums offer an online space for children to socialize with their friends, multiple reports identify such virtual playgrounds as a source of child predators. Short of banning screen time entirely, parents are hard-pressed to keep tabs on just what their children are doing online, and with whom. With online homework, this has become even more difficult. Even if this is an **ancillary** consequence of the underlying business model, the damage is done - children, from a tender age through adolescence, are becoming digitally addicted.

Similarly, right when children and youth are forming their initial views of the world, they are being sucked into virtual deep space, including the universe of fake news and conspiracy theories. **(III)** Many AI toys now come pre-programmed with their own **(A) amusing** and voice, and they may provide youngsters with **(B) personality** and creative options, with some even **(C) cognitive** improved reading, social skills, and **(D) fostering** development. All this is thrown at our children just when they are starting to try to make sense of who they are and the world they live in; right when it is so important to help them understand and appreciate different perspectives, preferences, beliefs and customs, to build bridges of understanding and empathy and goodwill.

46. The highlighted word in part (I) of the first paragraph may or may not have been used correctly.

Choose the most important word from the given options which can be used to replace the highlighted word. If no replacement is required, then choose option (e) as your answer choice.

- (a) slumping (b) tumbling
(c) worsening (d) inflaming
(e) No replacement required

47. Choose the most appropriate antonym of the word 'PRESSING', as highlighted in the given passage.

- (a) pivotal (b) drastic (c) trivial
(d) critical (e) None of these

48. Choose the most appropriate word from the following options to fill in the blank (II), as highlighted in the given passage.

- (a) wishing (b) earnest
(c) reluctant (d) craving
(e) esurient

49. Choose the most appropriate synonym of the word 'ANCILLARY', as highlighted in the given passage.

- (a) subsidiary (b) auxiliary
(c) adjuvant (d) accessory
(e) All of these

50. In sentence (III) of the last paragraph, a few words have been highlighted and jumbled. You are required to choose the correct sequence of the highlighted words from the options given below.

- (a) CADDB (b) BADCB (c) DBAC
(d) ABDC (e) CBAD

Directions (51-55): Read the following passage and answer the questions given below. Some words are highlighted to help you answer some of the questions.

Tropical deforestation is one of the great challenges facing humanity. If we don't slow it down soon, the chances of avoiding disastrous climate change, species loss and the disruption of indigenous and traditional people's cultures grow slimmer. And this rapid deceleration will be possible only if the governments, farm sectors, rural communities, businesses and civil societies of tropical forest regions devise and implement effective strategies. In the long term, the goal of taming deforestation must be embedded in public policies, programs, business transactions and culture. To put it bluntly, political leaders who embrace the goal of taming deforestation must be electable. The jurisdictional approach can help to translate unilateral corporate deforestation pledges into collaborative agreements with local governments and farm sectors, building the political case for slowing deforestation.

In a jurisdictional approach, the unit of performance is a landscape with policy-relevant boundaries - political geographies such as a state, province, district or an entire nation. This is the unit of performance that can effectively bring together multi-stakeholders on the appropriate pathway to sustainable development. In Brazil - a shining example of a successful jurisdictional approach to slow tropical deforestation - progress has been possible because political forces were pushing in the right direction. The approach was jurisdictional because success was defined as a decline in deforestation for the entire Brazilian Amazon region and for all types of land use. The Government used command-and-control measures to slow deforestation, but it also broadened its political base by paving its all-weather motorways providing access to Pacific markets. It also attracted new low-carbon industries to set up shop through innovative incentive systems.

The stage is also set to attract private financial investments into new low-carbon industries and farmer support programs as the risks associated with rampant deforestation, illegal land-use practices and labor infractions are diminished. In many ways, a political case for slowing Amazon deforestation was handed to President Lula on a silver platter. He came into office in 2003 with a mandate to deepen Brazil's international credibility just as Amazon deforestation was skyrocketing, triggering international outrage and a diplomatic headache for the new president. It was in this favorable political context that Lula empowered his environment minister to draw on the most powerful branches of the government - including the army and central bank - to crack down on illegal deforestation and logging in the Amazon. These formal designations further weakened the land-grabbing schemes that were driving deforestation.

51. What consequence did the use of command-and-control mechanisms by the government have?
 (a) It resulted in environmental protection by halting deforestation.
 (b) It contributed to the establishment of new low-carbon industries.
 (c) It successfully brought individuals together on the correct course.
 (d) Only (a) and (b)
 (e) Only (b) and (c)
52. Choose the most appropriate antonym of the word 'EMBRACE', as highlighted in the given passage.
 (a) enclasp (b) repudiate
 (c) espouse (d) underpin
 (e) stimulate
53. What will be the repercussions if tropical deforestation continues unabated?
 (a) Unrestrained deforestation will eventually result in the extinction of species.
 (b) Climate change will have a high probability of being catastrophic.
 (c) There will be a breakdown in traditional cultures and people.
 (d) Only (a) and (c)
 (e) All of these
54. Choose the most appropriate synonym of the word 'OUTRAGE', as highlighted in the given passage.
 (a) reconcile (b) indignation
 (c) propitiate (d) appease
 (e) conciliate
55. What role will private financial investments in innovative low-carbon industries play in maintaining the ecological balance?
 (a) It aids in reducing the dangers associated with massive deforestation.
 (b) It facilitates the reduction of unlawful land-use operations.
 (c) It contributes to the reduction of international indignation.
 (d) Only (a) and (b)
 (e) Only (a) and (c)

Directions (56-59): In each of the following questions, a sentence with four highlighted words has been given. One of these highlighted words may or may not have been used unnecessarily in the given sentences. You are required to choose the word that is redundant and gives a superfluous meaning. If all of the highlighted words are essential for the contextual meaning of the sentence, then choose option (e) as your answer choice.

56. Water-conservation and water-harvesting **methods (A)** that do not require the **construction (B)** of a dam are fallacious **approaches (C)** that have not been **adequately (D)** examined in the management of water resources.
 (a) A (b) B (c) C
 (d) D (e) None of these
57. Regional **hospitals (A)** at the secondary level within the government sector are **essential (B)** in urban facilities to **ensure (C)** the needs of the population during **critical (D)** situations.
 (a) A (b) B (c) C
 (d) D (e) None of these
58. The central government has re-issued an **ordinance (A)** forming an air quality **control (B)** commission, which worries **concern (C)** about the procedure of re-issuing ordinances without getting them **ratified (D)** by the Parliament.
 (a) A (b) B (c) C
 (d) D (e) None of these
59. The world appears to have **gleaned (A)** little from the first wave, which **culminated (B)** last year and **decimated (C)** the country, while a more **ferocious (D)** second wave is ravaging the country.
 (a) A (b) B (c) C
 (d) D (e) None of these

Directions (60-63): In each of the following questions, a contextually meaningful sentence has been given. Three paragraphs have been placed next to each sentence, which may or may not be relevant to the context of the given sentence. You are required to select the most appropriate combination of paragraphs from which the given sentence can be inferred.

60. The reasonable path is to enable both economic output and rapid health response, as evidenced by the rate of infection spread.
 (i) His address to the nation, delivered amid a roaring storm of viral outbreaks, was intended to reinforce the message about the commitment of the government to reversing economic decline and expanding healthcare services.
 (ii) To be clear, the fact that this was formerly the case does not imply that government control of educational institutions must presently be maintained. However, the circumstances that existed at the time the government passed its first endowment statute have not changed significantly.
 (iii) Regulatory reform proponents often point to the state's attitude to other activities. However, an examination of the associated laws reveals that the government also has significant supervisory responsibility for the management of educational properties.
 (a) Only (i) (b) Only (i) and (iii)
 (c) Only (iii) (d) Only (i) and (ii)
 (e) Only (ii) and (iii)
61. There have been no further guidelines created to describe these concepts or specify their constituents in order to assist the appointments.
 (i) Various qualifications and constitutional provisions for the freedom of conscience and the right to freely proclaim and disseminate religious practice according to public order were established with these objectives in mind.
 (ii) Recruiting agencies are required by the Court and the Model Police Act to shortlist candidates based on length of service, range of experience, and a performance evaluation of the candidates over the previous ten years.
 (iii) Reforms in this area must include two elements. The first is that the government's selection and dismissal powers should be delegated to an independent oversight body, of which the government is a component.
 (a) Only (iii) (b) Only (i) and (iii)
 (c) Only (ii) (d) Only (i) and (ii)
 (e) Only (ii) and (iii)
62. Following a supply outage, oxygen levels in numerous hospitals in the city fell to dangerously low levels. After a massive intervention, supplies were restored by late evening.
 (i) According to the Indian Council of Medical Research, only a small percentage of those who were vaccinated tested positive. These outbreaks of illness did not, however, call into question the efficacy of immunization as the vaccines clearly protect against disease.
 (ii) Hospitals in the city may soon face oxygen shortages as well. The Centre has a control team that will settle disagreements between states over oxygen delivery. To enable the smooth passing of oxygen trucks, green corridors have been established.
 (iii) After a leak in a medical oxygen tank disrupted the availability at a hospital, several patients on life support suffocated. According to the State Health Minister, the leak was caused by a valve malfunction, and a high-level investigation has been ordered.

- (a) Only (iii) (b) Only (i) and (iii)
 (c) Only (i) and (ii) (d) Only (ii)
 (e) Only (ii) and (iii)

63. It has been brought to our attention that, despite the summer break, several private schools continue to use the regular online learning approach.
- (i) Farmers in Punjab and Haryana have begun receiving the minimum support price payment directly in their bank accounts for the first time under the Direct Benefit Transfer Scheme, with wheat harvesting in full swing.
- (ii) The World Press Freedom Index has placed India in 142nd place out of 180 countries. This is despite the fact that an index monitoring cell worked for a year under the guidance of the Cabinet Secretary to enhance the ranks.
- (iii) Considering many private schools are continuing online classes despite the decree declaring summer break, the Directorate issued a circular stating that all online learning activities must be discontinued during the summer vacation.
- (a) Only (iii) (b) Only (i) and (iii)
 (c) Only (i) and (ii) (d) Only (ii) and (iii)
 (e) (c) Only (ii)

Directions (64-67): In the following question, two separate sentences with a few highlighted words have been given. Answer the questions, based on the given sentences.

64. The highlighted words in the given sentences may or may not have been used in the correct sequence. You are required to choose the correct sequence of words to form meaningful sentences.
- (A) The first wave of the pandemic **attempts** (A) the world **revealed** (B) the vulnerabilities in the **global** (C) healthcare system.
- (B) Hospitals have been **replenish** (D) for nearly a year, and no **throughout** (E) have been made to **swamped** (F) medical oxygen supplies.
- (a) BCEFDA (b) FBECAD
 (c) EFBDAC (d) EBCFAD
 (e) CFEDBA
65. You are required to choose the most appropriate connector from the given options which can be used to connect both sentences to form a coherent paragraph.
- (a) even (b) yet
 (c) since (d) then
 (e) for
66. The highlighted words in the given sentences may or may not have been used in the correct sequence.
- You are required to choose the correct sequence of words to form meaningful sentences.
- A. There has been a **achieving** (A) of headlines this week in the build-up **structured** (B) by the US regarding whether India should **sector** (C) a net-zero emissions target.
- B. Decarbonizing the power **declare** (D), which is India's single largest source of greenhouse gas **emissions** (E), is a critical step towards **rush** (F) net-zero emissions.
- (a) FBDCEA (b) BDFEAC
 (c) DBFACE (d) CBFEAD
 (e) FDBECA
67. You are required to choose the most appropriate connector from the given options which can be used to connect both sentences to form a coherent paragraph.
- (a) despite (b) although
 (c) however (d) consequently
 (e) moreover

Directions (68-72): In the following paragraph, a few phrases have been highlighted, each of which has been numbered. These highlighted phrases may or may not have been used correctly in terms of contextual meaning and grammatical accuracy. Select the most appropriate phrase from the given options to replace the incorrect one. Choose option (e) as your answer choice if the highlighted phrases are grammatically correct and don't need to be replaced.

Enhancing the efficiency of electricity use **are an important complement** (68) to decarbonizing the electricity supply. Growing urbanization and uptake of electricity services **offer a good opportunity to** (69) shape energy consumption within buildings through proactive measures. Cooling needs are expected **to increase rapid with rising** (70) incomes and temperatures. Air conditioners, fans and refrigerators together consume about 60% of the electricity in households. Today, the average fan sold in the **market consumes more than twice** (71) what an efficient fan does, and an average refrigerator is about 35% more. India can leverage this **transition to as an opportunity too** (72) become a global leader in the production of clean appliances.

68. (a) is an important complement
 (b) are the important complement
 (c) is the importance complement
 (d) are an important complements
 (e) No replacement required
69. (a) offers a good opportunity to
 (b) offering good opportunity to
 (c) offers good opportunity for
 (d) offered good opportunity to
 (e) No replacement required
70. (a) for increase rapidly with rising
 (b) to increased rapid with rising
 (c) to increase rapidly with rising
 (d) to increase rapidly with risen
 (e) No replacement required
71. (a) market consume more than twice
 (b) market consumes much than twice
 (c) market consumed more than twice
 (d) market consume even than twice
 (e) No replacement required
72. (a) transition to as an opportunity for
 (b) transition too as an opportunity to
 (c) transition to be an opportunity into
 (d) transition too as an opportunity of
 (e) No replacement required

Directions (73-77): A grammatically correct and contextually meaningful statement has been given in each of the following questions. Five similar sentences have been placed next to each statement, one of which is contextually similar in meaning to the one given in the question. Choose the most appropriate sentence that conveys the same meaning as the given statement.

73. Decision-makers must settle down to give the vaccination campaign wings by ensuring maximal industrial production in order to attain optimum distribution.
- (a) In order to achieve minimal dispersion, decision-makers must prolong the vaccination campaign's wings by promoting industrial production.
- (b) Decision-makers must stimulate steel production to broaden the wings of the vaccination campaign in order to achieve maximum diffusion.

- (c) In order to accomplish optimal vaccination coverage, decision-makers must encourage the manufacturing of medical equipment.
- (d) In order to obtain optimum dispersion, decision-makers must settle down to give the inoculation project wings by ensuring optimal industrial output.
- (e) Decision-makers must calm down to give the infection operation wings by delivering exceptional factory production in addition to enhancing diffusion.

74. Long before the Constitution was enacted, the competence to regulate the functioning of religious bodies was perceived as an instance of sovereignty.

- (a) The failure to regulate the functioning of religious organizations was seen as an illustration of sovereignty long before the Constitution was enacted.
- (b) Long before the Constitution was repealed, the power to govern the operations of religious organizations was overlooked as an example of sovereignty.
- (c) Long before the Constitution was enacted, the ability to govern the functioning of religious organizations was seen as a symbol of sovereignty.
- (d) Long before the Constitution was ratified, the reluctance to control the running of religious organizations was considered an example of autonomy.
- (e) The power to regulate the functioning of religious organizations was hampered long before the Constitution was abrogated as an example of sovereignty.

75. Recent events that led to the dismissal of an officer have brought the process of appointing and replacing state officials back into the forefront.

- (a) The procedure of dismissing and replacing state officials has resurfaced as a result of recent events that resulted in the removal of an officer.
- (b) Recent events involving the induction of a state officer have reintroduced the procedure of dismissing and replacing state officials.
- (c) Recent incidents surrounding the accession of a state officer have reintroduced the procedure of substituting and discharging state officials.
- (d) As a result of recent occurrences that resulted in the expulsion of an officer, the system for terminating and removing state officials has resurfaced.
- (e) The procedure of appointing and transferring state officials has resurfaced as a result of recent events that resulted in the firing of an officer.

76. The idea of governments monitoring the operations of religious institutions appears to be repugnant to a democratic republic on the surface.

- (a) On the surface, the idea of governments overseeing religious institutions appears to be appealing to a democratic republic.
- (b) On the surface, the idea of governments supervising religious institutions appears to be incompatible with a democratic republic.
- (c) On the surface, the idea of governments overseeing religious institutions looks to be enticing to an autocratic nation.
- (d) The idea of governments supervising religious institutions looks to be tempting to a constitutional democracy on the surface.
- (e) On the surface, the idea of governments monitoring religious institutions appears to be intriguing to a constitutional democracy.

77. According to publicly available data, India falls short of the global average of prescriptions administered per one lakh population.

- (a) According to publicly available data, India surpasses the global average of prescriptions dispensed per one lakh population.
- (b) India eclipses the global average of pharmaceuticals provided per one lakh people, according to privately available data.
- (c) According to publicly available data, India misses the mark of the global average of prescriptions provided per one lakh population.
- (d) According to commercially existing evidence, India beats the global average of medications delivered per one lakh inhabitants.
- (e) According to generally available information, India outperforms the global average of medications provided per one lakh population.

Directions (78-80): In the following questions, two columns are given containing three sentences/phrases each. In the first column, sentences/phrases are A, B and C and in the second column, the sentences/phrases are D, E and F. A sentence/phrase from the first column may or may not connect with another sentence/phrase from the second column. Each question has five options, four of which display the sequence(s) in which the sentences/phrases can be joined to form a grammatically and contextually correct sentence. Choose the appropriate option. If none of the given options forms a correct sentence after combination, mark (e), i.e. "None of these" as your answer.

78.

Column I		Column II	
A.	Such a sector-by-sector approach, which can be developed for other sectors	D.	and also to allow it to heighten its vows as required by the Paris Agreement.
B.	They can demonstrate concrete, yet ambitious, domestic action that sets	E.	India must take internal measures to get on the road to net-zero emissions.
C.	This approach allows India to gracefully adapt its sectoral transition plans	F.	while closing concrete actions that are less dependent on uncertain balances.

- (a) Only A-E and B-D (b) Only B-F
- (c) Only A-F and B-E (d) Only C-E
- (e) None of these

79.

Column I		Column II	
A.	India may even consider committing to submit plausible pathways and	D.	by authorities to disclose a confidential plan to launch The Super League.
B.	The European game has been thrown into turmoil by the decision taken	E.	timelines to achieving net-zero emissions as part of its future pledges.
C.	Most of the clubs considered themselves to be public-spirited entities in	F.	this excessive use of a football team as a mere for-profit business is the crux.

- (a) Only A-E and B-D (b) Only C-E
- (c) Only A-F and B-E (d) Only B-F
- (e) None of these

80.

Column I		Column II	
A.	Meaningful competition among them was seen as a meritocratic exercise	D.	outfits are answerable more to investors and owners than actual supporters.
B.	With leagues increasingly awash with hedge fund money and handouts of	E.	workforce longevity is vital from the perspective of the company's credibility.
C.	The nation hoped that the central government would take corrective action	F.	after watching and experiencing the repercussions of a medical supply crisis.

- (a) Only A-D and B-F (b) Only C-F
(c) Only A-E and B-D (d) Only B-E
(e) None of these

DATA ANALYSIS AND INTERPRETATION

81. Decide the combination of which rows of column I and II are sufficient to answer the question.

If a person sold an article, which

- I. costs ₹ _____
II. at _____ % profit
III. on a discount percentage of _____.
Find the selling price of article.

Column I	Column II
(a)	2) P
	3) $\frac{P}{2}$
(b)	1) 250
	2) $P - 1$
(c)	Profit earned = 20%
(d)	1) 250

- (a) a, d (b) b, d
(c) a, c (d) b, c
(e) Neither of combination is sufficient

82. Solve both equations and form a new equation in variable 'z' (reduce to lowest possible factor) using roots of equation I and II as per instructions given below.

I. $x + 2 = \frac{28}{x-1}$
II. $\frac{3}{2y-1} - \frac{4}{3y-1} = 1$

What will be new equation if its roots are highest root of equation I and lowest root of equation II?

- (a) $8z^2 - 34z - 9 = 0$ (b) $z^2 - 5z = 0$
(c) $8z^2 - 20z + 9 = 0$ (d) $z^2 - 8z = 0$
(e) None of these

Directions (83-84): Given below are two number series. Series I is a missing series while series II is a wrong number series which follows pattern of series I.

- I. 10, 59, P, 349, 638, 999
II. 240, 244, 252, 278, 327, 448, 617

83. If 2 is added in successive terms of a series starting with the nearest prime of P, then find the 5th term of this new series.

- A. 187 B. 165
C. 189 D. 163
(a) either A or C (b) either B or C
(c) either B or D (d) either C or D
(e) either A or B

84. If 'y' is the wrong number of the series II, then find the value of '2y - 1'.

- (a) 493 (b) 561 (c) 505
(d) 511 (e) 501

85. Car 'A' travels for 5 hours to reach from point X to Y. Car 'B' travels at the speed of 80 kmph to reach from point Y to Z in 3 hours.

Distance travelled by car 'A' to cover point X to Y is $\left(\frac{1}{8}\right)^{\text{th}}$ more

than distance travelled by car 'B'. How much time (approximate) will car 'A' takes to reach from point X to Z?

- (a) 9 hours (b) 7.5 hours
(c) 5 hours (d) 12 hours
(e) 9.4 hours

86. A mixture contains milk and water in the ratio of 5 : 1. If 20% of mixture is taken out and equal amount of milk and water are added into mixture, then difference between milk and water in final mixture becomes 96 liters. Find the initial amount of mixture.

- (a) 120 liter (b) 180 liter
(c) 90 liter (d) 150 liter
(e) None of these

Directions (87-91): The table given below shows the data about track record of bike racing game, where 4 bikes A, B, C and D are competing to reach finishing line which is at a distance of 2400 m from their initial starting point.

- I. Drift refers to over starting intentionally done by driver.
II. While calculating average speed, do not consider time taken in drift/crash.
III. Points = 50 × drifts - 20 × crashes

Bikes	Drift	Time to reach finishing line (in sec)	Crash
A	6	180	3
B	7	120	4
C	5	150	5
D	8	100	7

87. Which bike earned maximum points at the end of the race?

- (a) bike B (b) bike C
(c) bike A (d) bike D
(e) bike A and D

88. In order to gain 30 more points than bike D, how many total drifts should bike A have made, if bike A's time and number of crashes remain same?

- (a) 5 (b) 3 (c) 10
(d) 8 (e) 7

89. Another bike named T, makes total number of drifts same as that of bike D and it still receives 0 points at the end. How many crashes must bike T have suffered?

- (a) 15 (b) 18 (c) 21
(d) 12 (e) 20

90. Speed of bike D is what percentage more than that of bike B?

- (a) 25% (b) $33\frac{1}{3}\%$ (c) 50%
(d) 20% (e) 15%

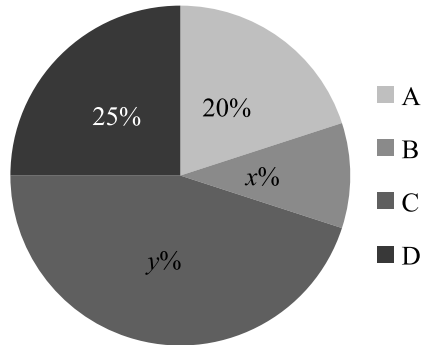
91. How much speed does bike C need to have in order to cover twice the distance in $\frac{5}{3}$ of the original time taken?
- (a) 16.8 m/s (b) 28 m/s
(c) 20.4 m/s (d) 19.2 m/s
(e) 10.5 m/s

Directions (92-96): The charts given below represents data of number of students of four colleges A, B, C and D. Based on the information given below, answer the questions that follow.

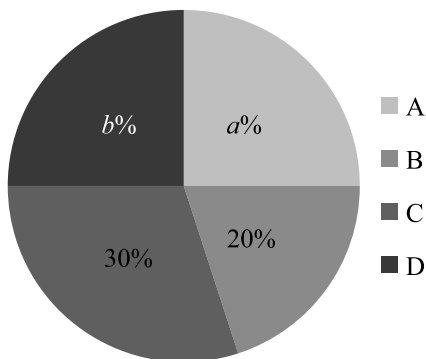
Notes:

- I. Number of girls in college B is 20.
II. Difference between number of boys and girls in college A is 200.
(Number of boys is greater than number of girls.)

Total number of students = 5000



Total number of boys = 2400



92. If 5% of boys from college D scored more than 85% in an exam, then find maximum number of boys who scored less than 85%.
(a) 440 (b) 400 (c) 570
(d) 250 (e) 320
93. Total number of boys in college D is what percent of total number of students in college A?
(a) 60% (b) 40% (c) 50%
(d) 80% (e) 20%
94. Find the difference between total number of girls in college A and C together and total number of boys in college B and D together.
(a) 575 (b) 775 (c) 925
(d) 850 (e) 650
95. Which of the following is true statement?
(a) Total number of students is least in college D.
(b) Total boys in college C is 720.
(c) Total girls in college A is 450.
(d) Difference between total number of boys in college A and that in college D is 80.
(e) None of these.

96. Find the ratio of boys in college B & C together and total girls in college C & D together.
(a) 60 : 117 (b) 60 : 103
(c) 60 : 113 (d) 60 : 107
(e) 60 : 109

Directions (97-102): Read the given below table carefully and answer the following questions.

The data shows volume and height of 4 right circular tanks P, Q, R and S. The given data also depicts time taken to fill or empty the tank by inlet pipe A, B and outlet pipe C.

- I. Time taken by pipe B and that by pipe C is different because pipes were used by the operator in different ways for different tanks.
II. Radius of tank S is 10.5 m.

Tank	Volume (in m ³)	Height (in m)	Time taken by pipe A to fill tank (in hours)	Ratio of time taken by pipe B to fill and pipe C to empty the tank
P	33264	24	10	2 : 3
Q	—	20	16	4 : 3
R	5390	35	18	5 : 6
S	19250	—	—	2 : 5

97. How much volume (approx.) of initially empty tank P was filled in 1 hour, when all 3 pipes were opened simultaneously? (Time taken by pipe C is 30 hours)
(a) 5235 m³ (b) 4528 m³ (c) 3881 m³
(d) 4865 m³ (e) 5743 m³
98. In tank R, pipe A was opened for 12 hours and then closed. Pipe B and C were the opened together for 3 hours and then both pipes were closed. After a total of 15 hours, what volume of tank remained empty (in cubic meters), (Note: if both pipes A and C were opened together, volume of tank will remain unchanged).
(a) 1775 m³ (b) 1841 m³ (c) 1617 m³
(d) 1029 m³ (e) 960 m³
99. Pipe B alone can fill the tank Q in 32 hours. When all the three pipes were opened for 1 hour, then volume of tank Q filled was approximately 856 cubic meter. What is the approximate radius of tank Q?
(a) 16 m (b) 21 m (c) 12 m
(d) 25 m (e) 9 m
100. When all 3 pipes were opened simultaneously, volume of tank S filled in 1 hour is 2575 cubic meter. Pipe B alone can fill 12.5% of tank S in 1.75 hours. In how much time can pipe A fill 40% of tank S alone.
(a) 6 hours (b) 4.8 hours
(c) 5.4 hours (d) 4.4 hours
(e) 3.3 hours
101. Find respective ratio of total surface area of tank P and curved surface area of tank R.
(a) 2 : 3 (b) 16 : 3 (c) 2 : 1
(d) 23 : 8 (e) None of these
102. Another tank Z has height 40% more than that of tank R and radius 4 m more than that of tank P. What is the sum of curved surface area of tank P and Z together?
(a) 9375 m² (b) 12348 m²
(c) 9935 m² (d) 7365 m²
(e) 10868 m²

Directions (103-105): Versha and Pinky started a business by investing ₹P and ₹1.2P respectively. After 'x' months, Versha withdrew her entire amount and Mamta entered into business. At the end of 9 months, Pinky increased her initial investment by 25%. On completion of one year, total share of Versha and Pinky in entire profit was ₹28950. If

Pinky would have increased her investment after 'x' months, then ratio of share of Versha and Pinky in entire profit would have been 5 : 21 respectively.

103. Find the value of 'x'.

- (a) 4 (b) 6 (c) 8
(d) 5 (e) 9

104. Find the difference between profit shares of Versha and Pinky.

- (a) ₹24350 (b) ₹5120 (c) ₹16950
(d) ₹17200 (e) ₹6700

105. If a man invested ₹1.5P for x years on simple interest at the rate of 4.5x % p.a. and received total interest of ₹8640, then find P.

- (a) 5400 (b) 8000 (c) 7200
(d) 6000 (e) 12000

106. B is 25% more efficient than A. If B was 50% more efficient than A, then B could complete the work in 15 days lesser than number of days taken by A. What fraction of work would be left after 9 days, if A and B work together?

- (a) $\frac{1}{5}$ (b) $\frac{7}{20}$ (c) $\frac{11}{20}$
(d) $\frac{13}{40}$ (e) $\frac{9}{20}$

107. A and B entered into a partnership and invested their respective amounts in the ratio of 5 : 3. C and D entered into another partnership with initial investment ratio of 4 : 5. Total investment of A and B was $33\frac{1}{3}\%$ less than that of C and D. If total profit of all 4 person at the end of year is ₹19800, then find profit of A.

- (a) ₹2000 (b) ₹4250
(c) ₹4950 (d) ₹4000
(e) ₹5600

Directions (108-110): Each questions below is followed by three statements I, II and III. You have to determine whether the data given in the statements are sufficient to answer the question. You have to choose the best possible answer.

108. The teacher distributes 'X' chocolates to 3n students. Find the value of n.

Statements:

- I. If he distributes 6 chocolates to each student, then he is left with 24 chocolates.
 - II. If the number of students were $66\frac{2}{3}\%$ less than the original number of students, then the teacher was left with no chocolates.
 - III. Number of students is more than 32 but less than 41 and number of chocolates is more than 210 but less than 270.
- (a) Data in statement I and II together is sufficient to answer the question.
(b) Data in statement I and III together is sufficient to answer the question.
(c) Data in statement II and III together is sufficient to answer the question.
(d) Data in statements I, II and III together are necessary to answer the question.
(e) Data in statements I, II and III together are not sufficient to answer the question.

109. P, Q and R entered into a partnership and after 8 months, Q and R left the business. What is their total annual profit at the end of the year?

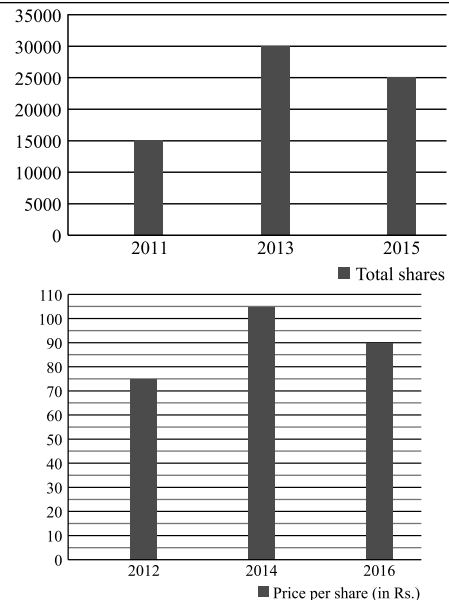
Statements:

- I. Annual profit of Q is ₹600 more than that of P and ₹300 more than that of R.
 - II. Amount invested by R is half of the total amount invested by P and Q together.
 - III. The ratio of the profit share of R and that of P and Q together is 1 : 2 respectively.
- (a) Data in statement I and II together is sufficient to answer the question.
(b) Data in statement I and III together is sufficient to answer the question.
(c) Data in statement II and III together is sufficient to answer the question.
(d) Data in statements I, II and III together is necessary to answer the question.
(e) Data in statements I, II and III together is not sufficient to answer the question.
110. Find the two-digit number.

Statements:

- I. Sum of the squares of the two digits of the two digits number is 26.
 - II. The ratio between the two-digit number and twice the sum of the digits of that number is 17 : 4.
 - III. The digit in ten's place is 4 more than the digit in unit place.
- (a) Any one of them
(b) Only I and II together are sufficient
(c) Either I and III together or II alone
(d) Only II and III together are s
(e) None of these

Directions (111-115): Read the data carefully and answer the questions.



Notes:

- (i) Market value = Total number of shares × price of one share
 - (ii) Number of shares increased by 20% from 2011 to 2012 and price of share decreased by 5% from 2012 to 2013.
 - (iii) Price of share decreased by 25% from 2015 to 2016 and number of shares decreased by 10% from 2015 to 2016.
111. In 2012, 80% of total market value received from old shareholders and rest received from new shareholders. Find the total market value received from new shareholders.
- (a) ₹240,000 (b) ₹350,000
(c) ₹22000 (d) ₹270,000
(e) ₹310,000

112. Find the ratio of total market value received in 2015 to that of in 2013.
 (a) 171 : 100 (b) 25 : 16 (c) 80 : 57
 (d) 171 : 250 (e) 250 : 151
113. Find total market value in 2016.
 (a) ₹202,5000 (b) ₹245,5000
 (c) ₹212,5000 (d) ₹239,5000
 (e) ₹199,5000
114. Find the percentage of increment in total market value in 2013 over 2012?
 (a) 37.5% (b) 132.5% (c) 65.5%
 (d) $58\frac{1}{3}\%$ (e) 27.5%
115. If total number shares in 2014 are 1870 less than average number of shares in 2013 and 2016, then find average number of shares from 2011 to 2016.
 (a) 18250 (b) 21250 (c) 22480
 (d) 23250 (e) 19250

GENERAL, ECONOMY/BANKING AWARENESS

116. The Report on Trend and Progress of Banking in India is a statutory publication in compliance with Section _____ of the Banking Regulation Act, 1949.
 (a) 58 (2) (b) 36 (2) (c) 47 (2)
 (d) 64 (2) (e) 31 (2)
117. According to AISTA, sugar mills have exported a total of 9,39,435 tonnes of sugar from which of the following period?
 (a) 1 April to 31 December 2021
 (b) 1 July to 31 December 2021
 (c) 1 July to 6 December 2021
 (d) 1 October to 31 December 2021
 (e) 1 October to 6 December 2021
118. For which of the following purposes ADB signed a USD 250 million loan pact with India?
 (a) Vaccines
 (b) Pradhan Mantri Gram Sadak Yojana
 (c) Mumbai-Ahmedabad Bullet Train Project
 (d) Infrastructure and Institutional Capacity Building
 (e) National Industrial Corridor Development Program
119. What is the public shareholding under NSE Prime which was enhanced by SEBI in December 2021?
 (a) 40% (b) 60% (c) 50%
 (d) 35% (e) 45%
120. Who among the following has been appointed as the new Chairperson of IN-SPACE?
 (a) Pawan Goenka (b) Anish Shah
 (c) S. Somnath (d) K. Sivan
 (e) Vikram Mehta
121. Which of the following partnered with Paytm Payments Bank to offer international fund transfers through full KYC-compliant Paytm wallets?
 (a) Muthoot Finance (b) Razorpay
 (c) Postpe (d) MoneyGram
 (e) Slice
122. The James Webb Space Telescope reached its final destination in December 2021. How much is the distance of the final destination of the telescope from Earth? (in million km)

- (a) 2 (b) 1 (c) 1.5
 (d) 2.5 (e) 3

123. The Reserve Bank of India has extended the validity of its directions for Punjab and Maharashtra Cooperative Bank from 1 January 2022 to _____.
 (a) 31 March 2022 (b) 31 December 2022
 (c) 30 June 2022 (d) 28 February 2022
 (e) 30 September 2022
124. Who among the following is the head of the Delimitation Commission?
 (a) Ranjana Prakash Desai (b) Ranjan Gogoi
 (c) PK Gupta (d) KK Sharma
 (e) Sushil Chandra
125. The Department of Atomic Energy announced in December 2021 to increase India's nuclear capacity from _____ MW to 22480 MW by 2031.
 (a) 12980 (b) 6780 (c) 8975
 (d) 13400 (e) 9240
126. The Archaeological Survey of India (ASI) acquired 84-acre of the land of the Bibi-ka-Maqbara, the historic monument replicating the famous Taj Mahal in December 2021. In which of the following states is the monument located?
 (a) Telangana (b) Maharashtra
 (c) Gujarat (d) Uttar Pradesh
 (e) Andhra Pradesh
127. When was the Union Budget presented on 1 February for the first time?
 (a) 2017 (b) 2021 (c) 2014
 (d) 2019 (e) 2015
128. Which of the following is not a component of the Pradhan Mantri Kisan Sampada Yojana (PMKSY)?
 (a) Food Safety & Quality Assurance Infrastructure
 (b) Mega Food Parks
 (c) Infrastructure for Agro-processing clusters
 (d) Operation Green
 (e) Creation of Backward & Forward Linkages
129. Which of the following methods was used by SBI & Union Bank for the NPA auctioning of KSK Mahanadi Power loans?
 (a) Swiss Challenge (b) Irish Challenge
 (c) Polish Challenge (d) French Challenge
 (e) German Challenge
130. In December 2021, it was announced to sell a 6% stake of _____ Mutual Funds through IPO.
 (a) ICICI (b) HDFC (c) SBI
 (d) UTI (e) Parag Parikh
131. In which of the following cities, Tansen Samaroh is held annually?
 (a) Indore (b) Gwalior (c) Devas
 (d) Jabalpur (e) Bhopal
132. Which of the following points is/are correct with respect to the central government's incentive scheme to promote RuPay debit cards and BHIM-UPI transactions?
 (a) The scheme applies to the banks having operations in India and transactions done in India.
 (b) The scheme is announced with an estimated financial outlay of Rs. 2000 crores.
 (c) The duration of the proposed scheme is one year, w.e.f. April 01, 2021.
 (d) Both 1 and 3
 (e) All 1, 2 and 3

133. Which of the following Asset Management Company (AMC) acquired a 100% stake in L&T Mutual Fund for a total cost of USD 425 million?
(a) HSBC (b) HDFC (c) IDFC
(d) IDBI (e) SBI
134. Who among the following is the author of the book "The Da Vinci Code"?
(a) Robert Frost (b) Ruskin Bond
(c) Jay Shetty (d) Dan Brown
(e) Chetan Bhagat
135. Which of the following states is not one of the top five performers in the overall performance in the large states category as per the 4th edition of the State Health Index 2019-2020?
(a) Karnataka (b) Kerala
(c) Tamil Nadu (d) Both 1 and 3
(e) All 1, 2 and 3
136. As per the updated SEBI guidelines, _____ portion of the applications will be reserved for Non-Institutional Investors with an application size of more than Rs 10 lakh rupees from April 2022.
(a) 2/3 (b) 1/3 (c) 1/4
(d) 2/5 (e) 1/2
137. Which of the following is not among the top five performers in the CFTI category in the 2021 ARIIA report?
(a) IIT Kanpur (b) IIT Bombay
(c) IIT Delhi (d) IIT Madras
(e) IIT Khadagpur
138. Which of the following organizations announced setting up a sophisticated semiconductor chip and glass manufacturing ecosystem in India?
(a) Vedanta (b) Zensar Technologies
(c) Alufouride (d) BDL
(e) Tata Motors
139. Which of the following banks launched Namaste UK accounts for Indian professionals on long-term visas to Britain?
(a) Bank of Baroda (b) ICICI Bank
(c) HSBC (d) State Bank of India
(e) Union Bank of India
140. What is the operational range of the Pralay missile?
(a) 150-250 km (b) 100-250 km
(c) 100-500 km (d) 250-500 km
(e) 150-500 km
141. DFC has approved \$500 million debt financing for First Solar's manufacturing facility in _____.
(a) Telangana (b) Karnataka
(c) Gujarat (d) Maharashtra
(e) Tamil Nadu
142. What was the rank of the Union Bank of India in the EASE 4.0 Reforms Index for Q1 FY22?
(a) third (b) fifth
(c) first (d) second
(e) fourth
143. The H-1B visa approval rate for fiscal 2021 was the highest in a decade. Which of the following countries issues the H-1B visa?
(a) Canada (b) Japan
(c) United States of America (d) United Kingdom
(e) South Korea
144. Sanjiv Mehta has been appointed as the President of _____.
(a) FSSAI (b) FICCI
(c) PETA India (d) Amul
(e) FCI
145. Which of the following banks has become the second largest merchant acquiring bank by installing over 2 lakh PoS machines in 2021?
(a) HDFC Bank (b) IndusInd Bank
(c) Kotak Mahindra Bank (d) Axis Bank
(e) ICICI Bank
146. What is the repayable term of the loans under the Stand-Up India scheme?
(a) 3 years (b) 12 years
(c) 10 years (d) 5 years
(e) 7 years
147. How many Lok Sabha sessions are conducted normally in a year?
(a) 1 (b) 5 (c) 4
(d) 2 (e) 3
148. How many households will be benefitted from the Gram Ujala programme that targets to distribute 10 million LED bulbs by 31 March 2022?
(a) 1 million (b) 10 million
(c) 5 million (d) 2 million
(e) 2.5 million
149. Who among the following chaired RBI's expert committee to review the governance of boards in India?
(a) P.J. Nayak (b) A.K. Bhuchar
(c) Ajay Shankar (d) Abid Hussain
(e) Deepak Mohanty
150. The first two digits in the newly introduced BH series number plate represent _____.
(a) state code (b) alphanumeric suffix
(c) BH code (d) registration year
(e) city code
151. Which of the following is India's first credit rating agency?
(a) CRISIL (b) CARE
(c) ICRA (d) India Ratings
(e) Acuite Ratings
152. Which of the following organizations has partnered to introduce a tokenization facility for Rupay card holders?
(a) SBI (b) SEBI
(c) NPCI (d) NABARD
(e) RBI
153. Which of the following became the first Indian bank to go live on UPI for cross-border payments and NRI remittances?
(a) ICICI Bank (b) IndusInd Bank
(c) Kotak Mahindra Bank (d) Punjab National Bank
(e) State Bank of India
154. SBI acquired a minority stake in JSW Cement for Rs 100 crore through CCPS. What does the first C in CCPS stand for?
(a) Combined (b) Conservation
(c) Capital (d) Convertible
(e) Compulsorily
155. What was the total financial outlay under the PLI scheme to promote semiconductor manufacturing in India?
(a) 76000 crore (b) 72000 crore
(c) 48000 crore (d) 88000 crore
(e) 64000 crore

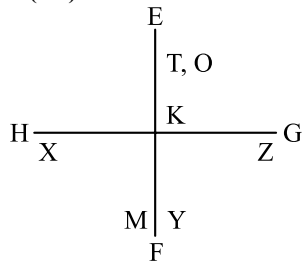


Answer Key & Solutions

1. (e) Both conclusions are valid. Conclusion I highlights the low priority given to school education, supported by the tardy progress in providing facilities. Conclusion II emphasizes the need for greater funding and prioritization of universal access and quality infrastructure in government schools.

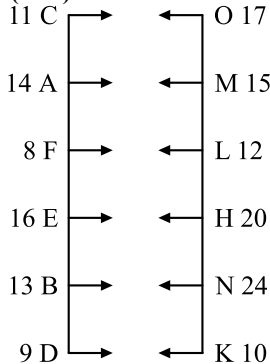
2. (a) As the statement highlights how rising delinquencies and tighter underwriting practices could limit refinancing options, adversely impacting existing borrowers. The other options either introduce unsupported details or fail to align logically with the provided information.

Solutions (3-5):



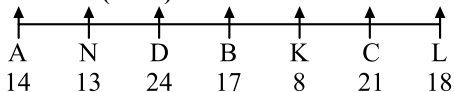
3. (d) Both (a) and (b)
4. (e) GO does not belong to that group.
5. (a) North East

Solutions (6-10):



6. (a) The building which has 12 offices
7. (c) Only I and II
8. (b) 11 is the average number of offices in buildings L, K, and C.
9. (e) All the above statements are true.
10. (d) Four buildings have more than 15 offices.

Solutions (11-15):



11. (c) The one who has 21 years of experience
12. (a) Only I and II

13. (b) L person sits at the right end of the row

14. (e) Two persons sit between B and the one who has 14 years of experience.

15. (d) Second to the left

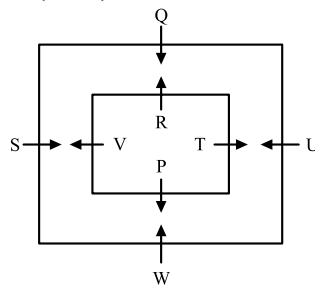
Solutions (16-20):

Biotech	Electronic	Mechanical
M(Silver), Q(Navy), T(Teal), P(Blue)	N(White), O(Maroon), R(Golden)	S(Pink), U(Green)

16. (b) Only III
17. (d) Pink color is liked by the one who works with U.
18. (a) Only II
19. (c) T - Green
All the pair of the persons are working in the same department except option (c)

20. (b) The one who likes Green

Solutions (21-23):



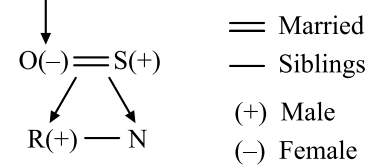
21. (b) Both II and III
22. (e) PW
23. (e) QR

Solutions (24-28):

Floor	Flat X	Flat Y
5	Pinaki	Arnab
4	Rahul	Mita
3	Iman	Bablu
2	Chandra	Umesh
1	Olivia	Gita

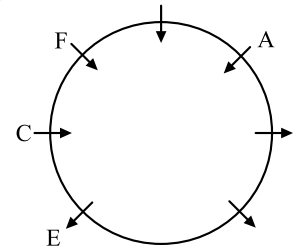
24. (d) The one who lives immediately above Umesh.
25. (a) Umesh's spouse
26. (a) North-east
27. (e) The one who is the spouse of Iman
28. (e) Floor 1, Flat X

29. (a)
P(+) = Q(-)



Both statements I and II together is sufficient to answer the question.

30. (b)



Both statements II and III together is sufficient to answer the question.

Solutions (31-34):

Input: Identify Eat 43 16 Actual 24 like 37 20 Gears However 39

Step I: Eat 17 Identify 43 Actual 24 like 37 20 Gears However 39

Step II: Eat 17 like 21 Identify 43 Actual 24 37 Gears However 39

Step III: Eat 17 like 21 Gears 25 Identify 43 Actual 37 However 39

Step IV: Eat 17 like 21 Gears 25 Actual 36 Identify 43 However 39

Step V: Eat 17 like 21 Gears 25 Actual 36 However 38 Identify 43

Step VI: Eat 17 like 21 Gears 25 Actual 36 However 38 Identify 42

31. (b) Four

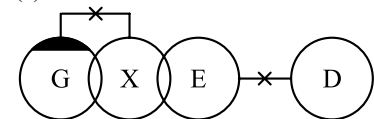
32. (d) 116

33. (c) Eat 17 like 21 Gears 25 Actual 36 Identify 43 However 39

34. (c) Gears

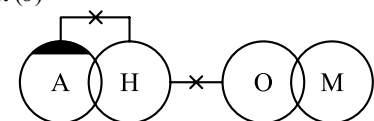
Solutions (35-37):

35. (c)



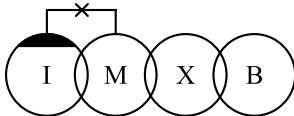
Both I and III

36. (b)



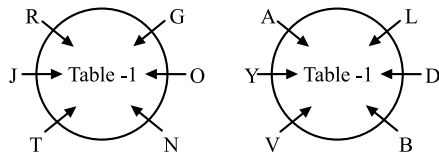
Both II and IV

37. (d)



Both I and II

Solutions (38-42):



38. (b) The one who sits to the left of J.

39. (a) The one who sits to the immediate left of R in table I

40. (c) The one who sits to the immediate left of B.

41. (e) Immediate left

42. (b) V sits second to the left of D in Table 2

43. (e)

(a) $C > N$ ($C \leq B = O = E = N$) → False

(b) $A \leq S$ ($A < C \leq B = O = E = N \leq S$) → False

(c) $S > O$ ($O = E = N \leq S$) → False

(d) $C < E$ ($C \leq B = O = E$) → False

(e) $C \leq S$ ($C \leq B = O = E = N \leq S$) → True

44. (b)

(a) $A = C$ ($A > V \geq U \geq B \geq C$) → False

(b) $V \geq C$ ($V \geq U \geq B \geq C$) → True

(c) $P > U$ ($U \geq B > P$) → False

(d) $K \geq P$ ($P < B \geq C > K$) → False

(e) $A \geq P$ ($A > V \geq U \geq B > P$) → False

45. (c) Three persons stand between F and A.

46. (e) The highlighted word in part (I) of the first paragraph is "revamping," which means "to give a new and improved form." It has been used correctly in the context of the passage. Thus, no replacement is required.

Slumping: Experiencing a sudden and significant decline or decrease.

Tumbling: Falling or dropping rapidly and uncontrollably.

Worsening: Becoming worse or deteriorating in quality or condition.

Inflaming: Causing something to become more intense, violent, or severe, especially in terms of emotions or conflicts.

47. (c) The word "pressing" in the passage means requiring quick or immediate action or attention. The most appropriate antonym for "pressing" would be something of little value or importance.

Trivial: Of little importance or significance; insignificant

Pivotal: Of crucial importance; central or essential to something.

Drastic: Extreme or severe; having a strong and far-reaching effect.

Critical: Of vital importance or essential significance; crucial or decisive.

48. (c) The word that best fits in the blank (II) is "reluctant". The sentence talks about parents being hesitant to allow their children to be photographed by the media.

Reluctant: Unwilling or hesitant; disinclined to do something.

Wishing: Expressing a desire for something to happen or be the case.

Earnest: Sincere and serious in intention, purpose, or effort.

Craving: Having an intense desire for something; longing for or yearning for intensely.

Esurient: Extremely hungry; voracious or greedy for food.

49. (e) The word "ancillary" in the passage means in addition to something else but not as important. Synonyms for "ancillary" include subsidiary, auxiliary, adjuvant, and accessory.

Subsidiary: Acting to assist or supplement; additional.

Auxiliary: Providing supplementary or additional help and support.

Adjuvant: Serving to aid or contribute; auxiliary.

Accessory: Contributing to or aiding an activity or process in a secondary way; supplementary.

50. (b) Revised sentence: Many AI toys now come pre-programmed with their own personality and voice, and they may provide youngsters with amusing and creative options, with some even fostering improved reading, social skills, and cognitive development.

"amusing" should be interchanged with "personality," and "cognitive" should be replaced with "fostering."

51. (d) The passage indicates that the government's use of command-and-control measures to slow deforestation had two significant consequences:

(a) It resulted in environmental protection by halting deforestation.

(b) It contributed to the establishment of new low-carbon industries.

Therefore, options (a) and (b) are both correct with reference to the context of the given question. Statement (c) is incoherent and not directly supported by the passage. The passage does not mention bringing individuals together on the correct course as a consequence of the government's actions.

52. (b) The most appropriate antonym of the word 'EMBRACE' is 'repudiate'

Repudiate: To reject or disown; to refuse to accept or be associated with.

Enclasp: To hold or grip tightly; to embrace.

Esouse: To adopt or support a cause, belief, or way of life.

Underpin: To provide support, strength, or a foundation for something.

Stimulate: To encourage or arouse interest, activity, or growth; to excite.

53. (e) By referring to the starting lines of the first paragraph, we can deduce that if tropical deforestation continues unabated, all of the given statements are correct with reference to the context of the passage. The passage highlights the disastrous consequences of unchecked deforestation, including the extinction of species, catastrophic climate change, and the breakdown of traditional cultures and communities. Therefore, option (e) "All of these" is the correct answer.

54. (b) The most appropriate synonym of the word 'OUTRAGE' is 'indignation',

Indignation: Anger or annoyance provoked by what is perceived as unfair treatment or injustice.

Reconcile: To restore friendly relations between; to make compatible or consistent.

Propitiate: To appease or pacify (someone or a deity) by performing a favor or making a gesture of goodwill.

Appease: To pacify or soothe (someone's anger or feelings); to relieve or satisfy (a demand or a need).

Conciliate: To stop someone from being angry or discontented; to placate or reconcile differences.

55. (d) Private financial investments in innovative low-carbon industries play a role in reducing the dangers associated with massive deforestation and facilitating the reduction of unlawful land-use operations.

56. (d) Emphasizes the examination's thoroughness, which is already implied by "examined." Removing "adequately" doesn't change the meaning.

57. (e) All words are necessary. "Regional" specifies the hospital's scope, "secondary level" clarifies its function within the healthcare system, "government sector" defines ownership, "urban facilities" indicate location, and "critical situations" highlight the importance of these hospitals.

58. (c) "Worries" and "concern" mean the same here. Removing "concern" makes the sentence more concise.

59. (e) All words are necessary. "Gleaned" implies learning something (even if little), "culminated" indicates the peak of the wave, "decimated" signifies the severity, "ferocious" describes the intensity of the second wave, and "ravaging" emphasizes its destructive nature.

60. (a) The sentence states that enabling both economic output and rapid health response is a reasonable approach, considering the infection rate.

- (i) This passage talks about the government's commitment to reversing economic decline and expanding healthcare services. It directly aligns with the concept of enabling both economic output and rapid health response.
- (ii) This passage discusses government control of educational institutions and doesn't relate to economic output or health response.
- (iii) This passage mentions regulatory reforms, but the context isn't related to the economic or health aspects mentioned in the statement.
- Based on the analysis, only passage (i) supports the statement. So the correct answer is (a) Only (i).
61. (e) Only paragraphs (ii) and (iii) are pertinent to the given sentence as they discuss recruiting, shortlisting, selection, and dismissal, all of which are relevant to appointments. The other paragraph does not align coherently with the context of the given sentence.
62. (e) After thoroughly examining the provided paragraphs, it is evident that only paragraphs (ii) and (iii) are applicable to the context of the given sentence. They discuss medical oxygen shortages, which directly align with the subject matter of the given sentence. In contrast, the other paragraph lacks coherence with the context of the given sentence.
63. (a) The provided sentence addresses the continuation of regular online classes despite the summer break. Accordingly, only paragraph (iii) is pertinent to the context of the given sentence as it also discusses the continuation of online classes despite the summer break order. Conversely, the other paragraphs are irrelevant to the context of the given sentence as they delve into unrelated topics.
64. (d) **Sentence A:** "The first wave of the pandemic revealed (B) the vulnerabilities in the global (C) healthcare system." - This order makes sense.
- Sentence B:** "Hospitals have been throughout (E) replenished (D) for nearly a year, and no supplies of medical oxygen have been made to swamped (F) medical oxygen supplies." - Here, "throughout" should come after "replenished" and "swamped" modifies "medical oxygen supplies."
65. (b) The first sentence talks about the pandemic revealing vulnerabilities, and the second sentence talks about hospitals being replenished. "Yet" connects these contrasting ideas by highlighting that despite the revelation of vulnerabilities, hospitals haven't been fully prepared.

66. (a) There has been a rush of headlines this week in the build-up structured by the US regarding whether India should declare a net-zero emissions target.
- Decarbonizing the power sector, which is India's single largest source of greenhouse gas emissions, is a critical step towards achieving net-zero emissions.
67. (d) The first sentence talks about the US pressuring India on net-zero targets. The second sentence emphasizes the importance of decarbonizing the power sector in achieving this goal. "Consequently" highlights the logical connection between the pressure and the critical step needed for success.
68. (a) "Is" agrees with the singular subject "Enhancing the efficiency..."
- "Complement" is the correct noun in this context, indicating something that adds to or improves decarbonization.
69. (a) "Offers" is the present tense verb that correctly reflects the ongoing opportunity. "To" introduces the infinitive phrase describing the action (shape energy consumption).
70. (c) "Increase rapidly" is a grammatically correct phrasal verb for fast growth.
71. (e) The phrase highlighted in the given sentence is grammatically and contextually correct. Thus, no replacement is required.
72. (b) Among the given options, only the phrase given in option (b) is correct to replace the highlighted one. Here, the adverb "too" should come first and the preposition "to" should come after to form a grammatically correct sentence
73. (d) This option uses "inoculation project" as a synonym for "vaccination campaign" and "optimal industrial output" for "maximal industrial production," maintaining the original meaning.
74. (c) This option keeps the core concept of "competence to regulate religious bodies" and "sovereignty" while using "symbol" as a synonym for "instance."
75. (e) This sentence accurately reflects the idea that recent events have brought attention back to the process of appointing and replacing state officials due to the dismissal of an officer.
76. (b) This option uses "incompatible with" to convey the idea of "repugnant to" and maintains the context of "democratic republic."
77. (c) This option uses "misses the mark of" as a synonym for "falls short of" and keeps "global average" and "prescriptions."
78. (e) None of the options create a fully coherent and grammatically correct sentence after combining phrases from both columns.
79. (a) If we combine options A-E and B-D, we can form two sentences:

- (i) India may even consider committing to submit plausible pathways and timelines to achieving net-zero emissions as part of its future pledges.
- (ii) The European game has been thrown into turmoil by the decision taken by authorities to disclose a confidential plan to launch The Super League.
80. (e) Out of the provided options, only the combination of C-F can form a grammatically correct and contextually meaningful sentence. The resulting sentence would be: "The nation anticipated that the central government would implement corrective measures following observation and firsthand experience of the aftermath of a medical supply crisis."
81. (d) Only from (b) and (c), selling price of article can be found.
- Cost price of article = ₹250
Profit % = 20%
So, selling price = 250 + 20% of 250
= 250 + 50 = ₹300
82. (b) I. $x + 2 = \frac{28}{x-1}$
 $\Rightarrow x^2 + x - 30 = 0 \Rightarrow x^2 + 6x - 5x - 30 = 0$
 $\Rightarrow (x-5)(x+6) = 0 \Rightarrow x = 5, -6$
 II. $\frac{3}{2y-1} - \frac{4}{3y-1} = 1$
 $\Rightarrow y + 1 = 6y^2 - 5y + 1 \Rightarrow 6y^2 - 6y = 0$
 $\Rightarrow 6y(y-1) = 0 \Rightarrow y = 0, 1$
 New eq. : $z^2 - sz + p = 0$
 Where, s = sum of roots, p = product of roots
 So, highest root of equation I and lowest root of equation II are 5 and 0 respectively.
 Required equation
 $= z^2 - (5+0)z + 5 \times 0 = 0$
 $\Rightarrow z^2 - 5z = 0$
83. (a) The pattern of the series I is:
 $10 + 7^2 = 59$
 $59 + 11^2 = 180 = P$
 $180 + 13^2 = 349$
 $349 + 17^2 = 638$
 $638 + 19^2 = 999$
 Prime numbers near P are 179 and 181.
 So, 5th term of the new series
 $= 179 + 2 + 2 + 2 + 2 = 187$
 Or, $181 + 2 + 2 + 2 + 2 = 189$
84. (c) Pattern of the series II:
 $240 + 2^2 = 244$
 $244 + 3^2 = 253$
 $253 + 5^2 = 278$
 $278 + 7^2 = 327$
 $327 + 11^2 = 448$
 $448 + 13^2 = 617$
 So, wrong no. is 253 = y
 So, required value = $253 \times 2 - 1 = 505$

85. (e) Distance travelled by car B from Y to Z
 $= 80 \times 3 = 240 \text{ km}$
 Distance travelled by car A from point X to Y

$$= 240 \times \frac{9}{8} = 270 \text{ km}$$

$$\text{Speed of car A} = \frac{270}{5} = 54 \text{ kmph}$$

$$\text{Required time} = \frac{240 + 270}{54}$$

$$= \frac{510}{54} = 9.4 \text{ hrs}$$

86. (b) Let total quantity of mixture be $30x$.

$$\text{Quantity of milk} = \frac{5}{6} \times 30x = 25x$$

$$\text{Quantity of water} = 30x - 25x = 5x$$

So, quantity of milk after 20% mixture taken out $= 80\%$ of $25x = 20x$

Quantity of water after 20% mixture taken out $= 80\%$ of $5x = 4x$

Let quantity of milk and water added be 'y'.

According to question,

$$(20x + y) - (4x + y) = 96$$

$$\Rightarrow 16x = 96 \Rightarrow x = 6$$

$$\text{Initial amount of mixture} = 6 \times 30$$

$$= 180 \text{ liter}$$

87. (a) Points of bike A $= 50 \times 6 - 20 \times 3 = 240$

$$\text{Points of bike B} = 50 \times 7 - 20 \times 4 = 270$$

$$\text{Points of bike C} = 50 \times 5 - 20 \times 5 = 150$$

$$\text{Point of bike D} = 50 \times 8 - 20 \times 7 = 260$$

So, earned maximum points by bike B.

88. (e) Points earned by bikes D

$$= 50 \times 8 - 20 \times 7 = 260$$

Let no. of drifts made by bikes A be N.

According to question,

Points earned by bike A

$$= 50 \times N - 20 \times 3 = 260 + 30$$

$$\Rightarrow 50N = 350 \Rightarrow N = 7$$

89. (e) Let number of crashes suffered by bike T be M.

According to question,

$$50 \times 8 - 20 \times M = 0$$

$$\Rightarrow M = \frac{400}{20} = 20$$

90. (d) Speed of bike D $= \frac{2400}{100} = 24 \text{ m/s}$

$$\text{Speed of bike B} = \frac{2400}{120} = 20 \text{ m/s}$$

So, required percentage

$$= \frac{24 - 20}{20} \times 100 = 20\%$$

91. (d) Required speed

$$= \frac{2 \times 2400}{\frac{5}{3} \times 150} = 19.2 \text{ m/s}$$

Solutions (92-96):

Number of students in college B

$$= \frac{x}{100} \times 5000 = 50x$$

Number of boys in college B

$$= \frac{20}{100} \times 2400 = 480$$

Number of girls in college B $= 50x - 480 = 20$

$$\Rightarrow x = 10$$

Total number of students in all the colleges together $= 20\% + 10\% + y\% + 25\% = 100\%$

$$\Rightarrow y = 45\%$$

Number of students in college A

$$= \frac{20}{100} \times 5000 = 1000$$

Number of boys in college A

$$= \frac{a}{100} \times 2400 = 24a$$

Difference between no. of boys and girls in college A $= 24a - (1000 - 24a) = 200$

$$\Rightarrow 48a = 1200 \Rightarrow a = 25$$

Total number of boys in all the given colleges together $= 25\% + 20\% + 30\% + b\% = 100\%$

$$\Rightarrow b = 25\%$$

Number of boys in college A

$$= \frac{25}{100} \times 2400 = 600$$

Number of girls in college A $= 1000 - 600$

$$= 400$$

Similarly,

College	Total students	No. of boys	No. of girls
A	1000	600	400
B	500	480	20
C	2250	720	1530
D	1250	600	650

92. (c) No. of boys of college D who scored

$$\text{more than } 85\% = \frac{5}{100} \times 600 = 30$$

So, maximum number of boys of college D who scored less than 85%

$$= 600 - 30 = 570$$

93. (a) Required percentage $= \frac{600}{1000} \times 100$
 $= 60\%$

94. (d) Required difference
 $= 400 + 1530 - (480 + 600)$
 $= 1930 - 1080 = 850$

95. (b) From above table, only statement (b) is correct.

96. (e) Required ratio

$$= (480 + 720) : (1530 + 650)$$

$$= 1200 : 2180 = 60 : 109$$

97. (c) Time taken by pipe B to fill the tank

$$\text{alone} = 30 \times \frac{2}{3} = 20 \text{ hours}$$

Let total capacity of tank be 60 units.

So, efficiency of pipe A, pipe B and pipe C is 6, 3 and -2 units/hr respectively.

So, required volume

$$= \frac{6 + 3 - 2}{60} \times 33264 \approx 3881 \text{ m}^3$$

98. (c) When both pipes A and C were opened together, volume of tank will remain unchanged. So, efficiency of pipe A and pipe C is same.

Time taken by pipe C to empty tank R alone = 18 hours

Time taken by pipe B to fill tank R alone

$$= \frac{18}{6} \times 5 = 15 \text{ hours}$$

Let capacity of tank R be 90 units.

So, efficiency of pipe A, B and C be 5, 6 and -5 units/hr respectively.

$$\text{Tank filled} = 12 \times 5 + (6 - 5) \times 3$$

$$= 63 \text{ units}$$

$$\text{Remained tank} = \frac{90 - 63}{90} \times 5390 = 1617 \text{ m}^3$$

99. (a) Time taken by pipe C to empty tank Q

$$= 32 \times \frac{3}{4} = 24 \text{ hours}$$

Let capacity of tank Q = 96 units

So, efficiency of pipe A, B and C is 6, 3 and -4 units/hr respectively.

Tank filled by all the pipes in 1 hour

$$= (6 + 3 - 4) \text{ units} = 856 \text{ cubic meter}$$

$$1 \text{ unit} = \frac{856}{5} \text{ cubic meter}$$

So, total capacity of tank Q

$$= \frac{856}{5} \times 96 \text{ cubic meter}$$

So, required radius

$$= \sqrt{\frac{856}{5} \times 96 \times \frac{7}{22} \times \frac{1}{20}} \approx 16 \text{ m}$$

100. (d) $12.5\% = \frac{1}{8}$

Time taken by pipe B alone to fill the tank

$$S = 1.75 \times 8 = 14 \text{ hours}$$

Time taken by pipe C to empty the tank S

$$\text{alone} = \frac{14}{2} \times 5 = 35 \text{ hours}$$

Let time taken by pipe A to fill the tank S alone be T hours.

According to question,

$$\frac{1}{T} + \frac{1}{14} - \frac{1}{35} = \frac{2575}{19250} \Rightarrow T = 11 \text{ hours}$$

$$\text{So, required time} = 11 \times \frac{40}{100} = 4.4 \text{ hours}$$

101. (e) Radius of tank P

$$= \sqrt{33264 \times \frac{7}{22} \times \frac{1}{24}} = 21 \text{ m}$$

Radius of tank R

$$= \sqrt{5390 \times \frac{7}{22} \times \frac{1}{35}} = 7 \text{ m}$$

So, required ratio

$$= \frac{2\pi \times 21 \times (21 + 24)}{2\pi \times 7 \times 35} = 27 : 7$$

102. (e) Height of tank Z = $35 \times \frac{140}{100} = 49 \text{ m}$

Radius of tank P

$$= \sqrt{33264 \times \frac{7}{22} \times \frac{1}{24}} = 21 \text{ m}$$

Radius of tank Z = $21 + 4 = 25 \text{ m}$

Required sum

$$= 2 \times \frac{22}{7} \times 21 \times 24 + 2 \times \frac{22}{7} \times 25 \times 49 = 10868 \text{ m}^2$$

Solutions (103-105):

Let investment of Mamta be ₹Y.

Ratio of share of profit of Versha, Pinky and Mamta

$$= Px : 1.2P \times 9 + 1.5P \times 3 : Y \times (12 - x)$$

If Pinky increase her investment after x months, Then,

$$\frac{P \times x}{1.2P \times x + 1.5P \times (12 - x)} = \frac{5}{21} \Rightarrow x = 4$$

Ratio of profit shares of Versha and Pinky

$$= Px : (1.2P \times 9 + 1.5P \times 3)$$

$$= x : 15.3 = 4 : 15.3 = 40 : 153$$

103. (a) Value of $x = 4$

104. (c) Required difference

$$= \frac{28950}{40 + 153} \times (153 - 40) = ₹16950$$

105. (b) According to question,

$$1.5P \times 4 \times \frac{4.5 \times 4}{100} = 8640$$

$$\Rightarrow P = 8000$$

106. (c) Let time taken by A to complete work alone be $30x$ days.

So, time taken by B to complete the work

$$\text{alone} = \frac{4}{5} \times 30x = 24x \text{ days}$$

Time taken by B to complete the work alone (when B 50% more efficient than

$$A) = \frac{2}{3} \times 30x = 20x \text{ days}$$

$$\text{So, } 30x - 20x = 15$$

$$\Rightarrow x = 1.5$$

So, time taken by A and B (original efficiency) to complete the work alone is 45 and 36 days respectively.

Let total work be 180 units.

$\therefore$ Efficiency of A and B is 4 units/day and 5 units/day respectively.

Work done in 9 days = $9 \times (4 + 5)$

$$= 81 \text{ units}$$

Fraction of work left

$$= \frac{180 - 81}{180} = \frac{99}{180} = \frac{11}{20}$$

107. (c) Let investment of A and B be $5a$ and $3a$ respectively.

Let investment of C and D be $4b$ and $5b$ respectively.

According to question,

$$(5a + 3a) = \frac{2}{3} \times (4b + 5b)$$

$$\Rightarrow \frac{a}{b} = \frac{3}{4}$$

Let $a = 3x$ and $b = 4x$

So, investment of A, B, C and D is $15x$, $9x$, $16x$ and $20x$ respectively.

$$\text{So, profit share of A} = \frac{15x}{60x} \times 19800$$

$$= ₹4950$$

108. (d) Number of students = $3n$

Number of chocolates = X

Number of students is the multiple of 3.

From I:

$$3n \times 6 + 24 = X \Rightarrow 18n + 24 = X \quad \dots (i)$$

From II:

Number of chocolates each student get =

$$\frac{X}{3n \times \frac{1}{3}} = \frac{X}{n} \quad \dots (ii)$$

From III:

Possible value of number of students ($3n$) = 33, 36 and 39

Or, possible value of $n = 11, 12$ and 13

From I, II and III:

Let $n = 11$

$$\text{So, } 18n + 24 = X$$

$$\Rightarrow 18 \times 11 + 24 = X \Rightarrow X = 222$$

But 222 is not divisible by 11.

So, this cannot be answer.

Let $n = 12$

$$\text{So, } 18n + 24 = X$$

$$\Rightarrow 18 \times 12 + 24 = X \Rightarrow X = 240$$

240 is divisible by 12.

Let $n = 13$

$$\text{So, } 18n + 24 = X$$

$$\Rightarrow 18 \times 13 + 24 = X \Rightarrow X = 258$$

But 258 is not divisible by 13.

So, this cannot be answer.

So, $x = 12$

Thus, data in statements I, II and III together are necessary to answer the question.

109. (e) Ratio of time for which P, Q and R invested respectively = $12 : 8 : 8 = 3 : 2 : 2$

From I and II:

Let the profit share of P = ₹ x

Profit share of Q = ₹ $(x + 600)$

Profit share of R = $(x + 600 - 300)$

$$= ₹(x + 300)$$

The ratio of individual investments of P, Q and R is not given.

So, data in statement I and II together is not sufficient to answer the question.

From I and III:

$$\frac{x + 300}{x + x + 600} = \frac{1}{2} \Rightarrow 2x + 600 = 2x + 600$$

So, data in statement I and III together are not sufficient to answer the question.

From II and III:

Only ratios are given. So, data in statement II and III together is clearly not sufficient to answer the question.

Data in all the statements together is also not sufficient to answer the question.

110. (c) Let the two digit number be xy .

$$\text{From I: } x^2 + y^2 = 26$$

So, the number can be 15 or 51 as

$$1^2 + 5^2 = 26$$

$\therefore$ Statement I alone is not sufficient.

$$\text{From II: } \frac{10x + y}{2(x + y)} = \frac{17}{4}$$

$$\Rightarrow 20x + 2y = 17x + 17y$$

$$\Rightarrow 3x = 15y \Rightarrow x = 5y$$

Only possible value of $y = 1$

So, the number is 51.

$\therefore$ Statement II alone is sufficient.

From I and III: $x^2 + y^2 = 26$ and $x - y = 4$

$$\text{So, } x = 5 \text{ and } y = 1$$

$\therefore$ Statements I and III together are sufficient.

111. (d) Total shares in 2012

$$= 15000 \times \frac{120}{100} = 18000$$

Total market value received from new shareholders

$$= 18000 \times 75 \times \frac{20}{100} = ₹270,000$$

112. (c) Price per share in 2015

$$= 90 \times \frac{100}{75} = ₹120$$

Price per share in 2013

$$= 75 \times \frac{95}{100} = ₹\frac{285}{4}$$

Required ratio

$$= (25000 \times 120) : (30000 \times \frac{285}{4})$$

$$= 80 : 57$$

113. (a) Total number of shares in 2016

$$= 25000 \times \frac{90}{100} = 22500$$

Total market value in 2016
 $= 22500 \times 90 = ₹2025000$

114. (d) Total shares in 2012

$$= 15000 \times \frac{120}{100} = 18000$$

Total market value in 2012

$$= 18000 \times 75 = ₹1350000$$

Price per share in 2013

$$= 75 \times \frac{95}{100} = ₹\frac{285}{4}$$

Total market value in 2013

$$= \frac{285}{4} \times 30000 = ₹2137500$$

Required percentage

$$= \frac{21375000 - 1350000}{1350000} \times 100 = 58\frac{1}{3}\%$$

115. (c) Total shares in 2016

$$= 25000 \times \frac{90}{100} = 22500$$

Total number of shares in 2014

$$= \frac{22500 + 30000}{2} + 1870 = 24380$$

Required average

$$= \frac{15000 + 18000 + 30000 + 24380}{6}$$

$$= \frac{134880}{6} = 22480$$

116. (b) The Reserve Bank of India releases the Report on Trend and Progress of Banking in India which is a statutory publication in compliance with Section 36 (2) of the Banking Regulation Act, 1949.

The report was last released in December 2021.

This Report presents the performance of the banking sector, including co-operative banks, and non-banking financial institutions during 2020-21 and 2021-22.

117. (e) According to the All India Sugar Trade Association (AISTA), sugar mills have exported 9.39 lakh tonnes of the sweetener till the first week of December during the ongoing 2021-22 marketing year that started from October 1 till 6 December 2021.

During the marketing year 2020-21, the country exported a record 7.23 million tonnes of sweetener.

The sugar marketing year runs from October to September.

118. (e) In October 2021, the Asian Development Bank (ADB) approved a \$250 million policy-based loan to support India's National Industrial Corridor Development Program (NICDP).

This loan is the first subprogram of a \$500 million programmatic loan aimed at

developing 11 industrial corridors across 17 states in India.

The NICDP aims to enhance the competitiveness of India's manufacturing sector by developing world-class industrial nodes supported by efficient urban agglomerations and multimodal connectivity, including ports and airports.

119. (a) The Securities and Exchange Board of India (SEBI) has increased the public shareholding under NSE Prime from 25% to 40% in December 2021.

The National Stock Exchange (NSE) launched 'NSE Prime' in 2021 which is a framework that allows listed companies to sign-up for higher governance standards than what is required under the law.

Tuhin Kanta Pandey is the new chairman of capital markets regulator SEBI replacing Madhabi Puri Buch.

120. (a) Former managing director of Mahindra & Mahindra Pawan Kumar Goenka has been appointed as the new Chairperson of the Indian National Space Promotion Authorization Centre (IN-SPACe).

To facilitate private sector participation, the government created the Indian National Space Promotion and Authorisation Centre (IN-SPACe) in 2020 as a single-window, independent, nodal agency which functions as an autonomous agency in the Department of Space (DOS).

Prime Minister Narendra Modi inaugurated the headquarters of the Indian National Space Promotion and Authorisation Centre (IN-SPACe) at Bopal, Ahmedabad in June 2022.

121. (d) Paytm Payments Bank, an associate entity of Paytm, and Nasdaq-listed MoneyGram, a global leader in digital P2P payments, have entered into a partnership to enable MoneyGram customers globally to send money in real-time to a Paytm Wallet user in India.

With this partnership, MoneyGram users abroad can now conveniently transfer money to full KYC Paytm wallet users in India from the comfort of their homes.

This is MoneyGram's first mobile wallet partnership in India and is an important development as the country's evolution to digital receivers is occurring rapidly.

122. (c) The James Webb Space Telescope arrived at its cosmic parking spot which is 1.5 million km away, bringing it a step closer to its mission to unravel the mysteries of the Universe.

Webb began its science mission this summer using its high-resolution infrared instruments to peer back in time 13.5

billion years to the first generation of galaxies that formed after the Big Bang.

The James Webb Space Telescope, which is expected to cost Nasa nearly \$10 billion, is one of the most expensive scientific platforms ever built.

123. (a) The Reserve Bank of India (RBI) has extended the validity of its directions for the scam-hit Punjab and Maharashtra Cooperative (PMC) Bank from 1 January 2022 to 31 March 2022.

Unity Small Finance Bank paid around ₹3800 crore to depositors of the scam-hit PMC Bank after the Deposit Insurance and Credit Guarantee Corporation (DICGC) had certified about 850000 such valid accounts.

124. (a) Justice Ranjana Prakash Desai, a retired judge of the Supreme Court of India, was appointed as the Chairperson of the Delimitation Commission of India on 13 March 2020.

The Delimitation Commission is responsible for redrawing the boundaries of various assembly and parliamentary constituencies based on the latest census data.

Under her leadership, the Commission finalized the delimitation order for the Union Territory of Jammu and Kashmir, increasing the number of assembly constituencies from 83 to 90.

125. (b) The Department of Atomic Energy announced in December 2021 to increase India's nuclear capacity from 6780 MW to 22480 MW by 2031.

The government has accorded administrative approval and financial sanction for 10 indigenous Pressurised Heavy Water Reactors (PHWRs) with an aggregate capacity of 7000 MW to be set up in fleet mode.

126. (b) The Archaeological Survey of India (ASI) acquired 84-acre of the land of the Bibi-ka-Maqbara, the historic monument replicating the famous Taj Mahal in December 2021.

The monument is a mausoleum of Rabi-ul-Daurani, alias Dilras Banu Begum, who was the wife of the Mughal emperor Aurangazeb.

The monument is believed to be constructed by prince Azam Shah in memory of his mother between 1651 and 1661 AD in Aurangabad, Maharashtra.

127. (a) Arun Jaitley started presenting the Union Budget on 1 February 2017, departing from the colonial-era tradition of using the last working day of that month.

The Budget was first introduced in India on 7 April 1860 when Scottish economist and politician James Wilson from East

India Company presented it to the British Crown.

India's first budget was presented on 26 November 1947 by the then Finance Minister R K Shanmukham Chetty.

128. (d) SAMPADA (Scheme for Agro-Marine Processing and Development of Agro-Processing Clusters) was approved by the cabinet in May 2017.

The scheme was renamed Pradhan Mantri Kisan Sampada Yojana (PMKSY).

It is an umbrella scheme incorporating ongoing schemes of the ministry like Mega Food Parks Integrated Cold Chain and Value Addition Infrastructure Food Safety and Quality Assurance Infrastructure for Agro-processing Clusters Creation of Backward and Forward Linkages Creation / Expansion of Food Processing & Preservation Capacities.

129. (a) State Bank of India (SBI) and Union Bank of India used the Swiss Challenge auction method for the KSK Mahanadi Power loans.

In the Swiss auction process, lenders first hold an auction of the NPA, and the offer made by the highest bidder (also known as the anchor bidder) is set at a reserve price.

130. (c) The executive committee of the central board of the State Bank of India (SBI) has accorded approval for exploring possibilities to offload a 6% of the Bank in SBI Funds Management Private Limited through the IPO route, subject to receipt of all regulatory approvals.

SBI Mutual Funds is a joint venture between India's largest public-sector lender State Bank of India and France's Amundi Asset management.

Challa Sreenivasulu Setty is an Indian banker and current Chairman of the State Bank of India (SBI).

131. (b) Tansen Samaroh is a 5-day extravaganza celebrated every year in December in Gwalior district, Madhya Pradesh.

Artists and music lovers from all over the world gather here to pay tribute to the Great Indian Musical Maestro Tansen.

Mangubhai (c) Patel is the Governor of Madhya Pradesh.

132. (d) In December 2021, the union cabinet chaired by Prime Minister Narendra Modi approved an incentive scheme to promote RuPayDebit cards and low-value BHIM-UPI transactions (Person-to-Merchant (P2M)) in the country.

Under the scheme, the acquiring banks will be incentivised by the government, by way of paying a percentage of the value of transactions (P2M) done through RuPay Debit cards and low-value BHIM-UPI modes of payments, at an estimated

financial outlay of Rs.1300 crore for one year with effect from 1 April 2021.

The scheme applies to the banks having operations in India and transactions done in India.

133. (a) Foreign lender HSBC announced in December 2021 that its asset management arm will acquire L&T Mutual Fund for USD 425 million.

L&T Investment Management (LTIM), a wholly-owned subsidiary of LTFH, is the investment manager of L&T Mutual Fund.

134. (d) The book "The Da Vinci Code" is authored by Dan Brown, an American author known for his thriller novels that blend historical themes, cryptography, and conspiracy theories.

Published in 2003, the novel became a global bestseller, gaining immense popularity for its gripping plot and controversial interpretations of religious history. A movie was released with the same name in 2006.

135. (a) NITI Aayog released the fourth edition of the State Health Index for 2019–20.

The report, titled "Healthy States, Progressive India", ranks states and Union Territories on their year-on-year incremental performance in health outcomes as well as their overall status.

On overall ranking based on the composite index score in 2019–20, the top-ranking states were Kerala and Tamil Nadu among the 'Larger States', Mizoram and Tripura among the 'Smaller States'.

136. (a) The Securities and Exchange Board of India (SEBI) has tightened certain rules regarding initial public offerings.

These include new guidelines for determining quota for high net-worth individuals and a longer lock-in period for anchor investors

From 2022, 33% of shares allocated to non-institutional investors will be reserved for investors with application sizes ranging from Rs 2 lakh to less than Rs 10 lakh. The remaining, 2/3 of the portion, will be earmarked for applicants whose application size exceeds Rs 10 lakh.

137. (e) Union Minister of State for Education, Dr Subhas Sarkar virtually announced the Atal Ranking of Institutions on Innovation Achievements (ARIIA) 2021 in December 2021.

IIT Madras, IIT Bombay, IIT Delhi, IIT Kanpur and IIT Roorkee are the top five performers in the CFTI category in the 2021 ARIIA report.

138. (a) Vedanta has announced setting up a sophisticated semiconductor chip and glass manufacturing ecosystem in India.

It will invest up to Rs 60,000 crore to set up this ecosystem.

139. (d) The State Bank of India is celebrating its centenary year in the UK with the launch of a new 'Namaste UK' account, aimed at benefitting Indian students and professionals on long-term visas to be able to set up their banking services in Britain before they set off from India.

Additional plans are aimed at New to Bank (NTB) customers, allowing them to use SBI UK remittance facilities without the need to open a bank account.

140. (e) Defence Research and Development Organisation (DRDO) successfully conducted a maiden flight test of the indigenously developed surface-to-surface missile 'Pralay', from Dr A P J Abdul Kalam Island off the coast of Odisha on December 22, 2021.

The missile has a range of 150-500 kilometres and can be launched from a mobile launcher.

141. (e) Development Finance Corporation (DFC) has approved \$500 million in debt financing for First Solar's manufacturing facility in Tamil Nadu.

The facility has a projected annual capacity of 3.3 gigawatts (GW).

142. (a) Union Bank of India ranked third amongst twelve Public Sector Banks (PSBs) in the EASE 4.0 Reforms Index for Q1 FY22, as per the rank list released by Indian Banks' Association (IBA).

Union Bank of India has been adjudged winner in the themes 'Collaborating for synergistic outcomes' & 'Governance & outcome-centric HR' and bagged Runner-Up positions in 'New age 24x7 Banking' & 'Tech-enabled ease of Banking' themes.

Union Bank of India is headquartered in Mumbai, Maharashtra.

143. (c) The H-1B visa is issued by the United States of America.

This non-immigrant visa allows U.S. employers to hire foreign professionals in specialty occupations requiring theoretical or technical expertise.

In fiscal year 2021, the H-1B visa approval rate surged to 97%, marking the highest in a decade.

The H-1B program is particularly significant for Indian professionals, who constitute a substantial portion of its beneficiaries.

144. (b) Sanjiv Mehta, Chairman and Managing Director of Hindustan Unilever Limited (HUL) has been appointed as the Federation of Indian Chambers

of Commerce and Industry (FICCI) President.

Alongside HUL, Mehta is also a Director on the Board of the Indian School of Business, a member of the Breach Candy Hospital Trust, and the South Asia Advisory Board of Harvard Business School.

145. (d) Axis Bank has become the second-largest merchant-acquiring bank in the country, leapfrogging two spaces during 2021 by installing over two lakh card-swipe machines during the year.

This is part of the bank's 'Axis One' strategy where it approaches the customers by offering its entire gamut of products instead of a standalone service.

Amitabh Chaudhry is the MD & CEO of Axis Bank.

146. (e) The Stand-Up India scheme, launched by the Government of India, aims to promote entrepreneurship among Scheduled Caste (SC), Scheduled Tribe (ST), and women entrepreneurs by facilitating bank loans for setting up greenfield enterprises.

Under this scheme, the repayment term for loans is up to 7 years, with a maximum moratorium period of 18 months, allowing borrowers some flexibility before commencing repayments.

The scheme offers composite loans ranging from ₹10 lakh to ₹1 crore to support new ventures in manufacturing, services, trading, or agri-allied sectors.

147. (e) Lok Sabha sessions are conducted normally three times a year- Budget Session, Monsoon Session and Winter Session.

Lok Sabha is composed of representatives of the people chosen by direct election based on adult suffrage.

The maximum strength of the Lok Sabha is 552.

148. (d) The Gram Ujala programme, launched by Convergence Energy Services Limited (CESL), aims to distribute 10 million (1 crore) LED bulbs to rural households across India by 31 March 2022.

Each rural household is eligible to receive up to 5 LED bulbs, priced at ₹10 per bulb, in exchange for working incandescent bulbs.

Given the distribution cap of 5 bulbs per household, the programme is set to benefit approximately 2 million households.

149. (a) Dr P.J. Nayak chaired RBI's expert committee to review the governance of boards in India.

The Expert Committee was requested to examine the working of banks' boards including whether the adequate time is devoted to issues of strategy, growth, governance and risk management; to review central bank regulatory guidelines on bank ownership, ownership concentration and representation in the board.

150. (d) The Ministry of Road Transport & Highways (MoRTH) has introduced a new "BH" series (Bharat Series) number plate/ registration mark for vehicles.

The format of the registration number, and on the plate would feature the year of registration, the BH code, followed by an alphanumeric suffix with a series of four numbers and two letters.

151. (a) CRISIL is India's first credit rating agency, incorporated in 1987 and promoted by the erstwhile ICICI Ltd along with UTI and other financial institutions.

CRISIL commenced operations within a year of its incorporation on 1 January 1988.

152. (c) The National Payments Corporation of India (NPCI) has joined hands with brands and aggregators such as BigBasket, Goibibo, MakeMyTrip, JioPay, Paytm and PhonePe to introduce a tokenization facility for Rupay card users to enhance online transaction security.

The facility will be supported by the recently launched NPCI Tokenization System (NTS).

Tokenization will allow a seamless and convenient transaction experience with heightened data security.

153. (b) IndusInd Bank has joined hands with the National Payments Corporation of India (NPCI) for offering real-time cross-border remittances to India using UPI IDs, for its Money Transfer Operator (MTO) partners.

With this initiative, IndusInd Bank has become the first Indian bank to go live on UPI for Cross Border Payments/NRI Remittances.

154. (e) The State Bank of India (SBI) has acquired a minority stake in JSW Cement Limited, with an investment of Rs 100 crore in the company via Compulsorily Convertible Preference Shares (CCPS).

Part of the diversified \$13-billion JSW Group, JSW Cement has manufacturing units in Vijayanagar in Karnataka, Nandyal in Andhra Pradesh, Salboni in West Bengal, Jajpur in Odisha, and Dolvi in Maharashtra.

155. (a) The government had announced an outlay of Rs 76000 crore, under its PLI scheme, separately for the development of a semiconductor and display manufacturing electronics ecosystem.

India's semiconductor component market is likely to reach \$300 billion in cumulative revenues by 2026.

